



INFORMATICS  
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In Collaboration with

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## **SalesSense**

### **Machine Learning Based Sales Forecasting for Retails Shops**

A Project by

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### **ABSTRACT**

This project addresses the challenge of inaccurate sales forecasting in small-scale retail businesses, particularly in niche domains such as spice retailing, where poor predictions lead to inefficient inventory management and suboptimal decision-making. The study proposes a machine learning-based approach using the XGBoost regression model to generate accurate six-month sales forecasts from historical data. To improve model reliability, time-based and lag features are engineered to capture seasonal patterns and temporal dependencies. The system further incorporates SHAP-based interpretability to explain model predictions and enhance transparency for non-technical users. In addition, a scenario-based component is included to allow users to adjust business parameters and explore different forecasting outcomes. The results demonstrate that the proposed approach provides a practical and interpretable solution for supporting inventory planning and data-driven decision-making in small retail environments.

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## LIST OF ABBREVIATIONS

| Abbreviation | Definition                       |
|--------------|----------------------------------|
| ML           | Machine Learning                 |
| DL           | Deep Learning                    |
| LSTM         | Long Short-Term Memory           |
| XGBoost      | Extreme Gradient Boosting        |
| SME          | Small and Medium Enterprises     |
| SRS          | System Requirement Specification |
| MAE          | Mean Absolute Error              |
| RMSE         | Root Mean Square Error           |
| MAPE         | Mean Absolute Percentage Error   |
| SHAP         | SHapley Additive Explanations    |

## CHAPTER 01: INTRODUCTION

### 1.1 Chapter Overview

This chapter provides an introduction to the research project, focusing on the development of a machine learning–based sales forecasting system for a small-scale retail spice business. The chapter begins by outlining the real-world challenges faced by small retailers, particularly in demand prediction and inventory decision-making. It then defines the research problem and highlights the limitations of traditional forecasting approaches. The chapter further presents the motivation for adopting advanced machine learning techniques, specifically the use of the XGBoost regression model to generate accurate six-month sales forecasts. In addition to prediction, the system incorporates scenario-based analysis, allowing users to adjust business parameters such as cost assumptions to support decision-making. To enhance transparency, SHAP-based interpretability is integrated to explain model predictions. Finally, the chapter defines the research aim, objectives, scope, and key contributions of the study, establishing a clear foundation for the subsequent chapters.

### 1.2 Problem Background

#### 1.2.1 Challenges Faced by Small Retail Businesses

Small retail businesses, particularly those operating in sectors such as specialty food ingredients and spices, face a dynamic and unpredictable environment. One of the primary challenges is aligning critical operational functions such as inventory management, financial planning, and resource allocation with ever-changing consumer demand.

- Inaccurate demand forecasting often leads to two costly outcomes: Overstocking, which ties up capital in slow-moving or perishable goods and increases holding costs.
- Understocking, which results in missed sales opportunities, reduced customer satisfaction, and potential long-term brand damage.

For a specialty spice shop, these challenges are intensified by a variety of influencing factors including seasonal demand, cultural festivals, holiday cooking trends, and promotional activities. Traditional forecasting methods that rely on simple averages or basic regression fail to capture these complex, non-linear market dynamics. Consequently, small business owners are often forced to make reactive decisions rather than proactive, data-driven strategies.

### 1.2.2 Limitations of Traditional Forecasting Techniques

Historically, sales forecasting in small and medium-sized enterprises (SMEs) has relied heavily on basic statistical models such as Linear Regression or Moving Averages. While these techniques provide a general understanding of trends, they often underperform in real-world retail contexts where sales data is non-linear, high-dimensional, and influenced by multiple factors.

Studies have shown that such models frequently yield low explanatory power, indicating their limited ability to accurately represent complex patterns in retail data (Ganguly & Mukherjee, 2024). As a result, businesses relying solely on traditional methods suffer from inefficient inventory control, budget mismanagement, and financial uncertainty.

The rapid evolution of digital commerce, coupled with an abundance of transactional and financial data, highlights the urgent need for advanced analytical methods capable of handling these intricate relationships.

### 1.2.3 The Need for Machine Learning-Based Forecasting

The advancement of machine learning (ML) has significantly improved the capability of forecasting systems to handle complex and non-linear data patterns. Unlike traditional statistical approaches, ML models such as Random Forest, XGBoost, and Long Short-Term Memory (LSTM) are capable of capturing hidden relationships within time series data, making them more suitable for retail sales prediction.

For a small-scale spice retail business, the application of ML enables more accurate multi-step-ahead forecasting, particularly for predicting sales over a six-month horizon. These predictions provide valuable support for key business decisions, including inventory planning and demand management.

In this project, the forecasting output is further extended through a scenario-based approach, where users can adjust business parameters such as estimated cost or demand variations. This allows the system to provide flexible, what-if analysis rather than fixed financial predictions.

By integrating machine learning-based forecasting with user-driven inputs and interpretable outputs, small business owners can shift from reactive decision-making to more informed and data-driven strategies.

### 1.3 Problem Definition

The core problem addressed in this research is the lack of a reliable and data-driven forecasting solution for small-scale retail businesses, particularly in niche domains such as spice retailing. Many small retailers rely on manual estimation or basic statistical methods, which fail to capture seasonal and non-linear sales patterns. As a result, decisions related to inventory and demand planning are often made without an accurate forward-looking perspective. Therefore, the problem lies in the need for an accurate, simple, and interpretable system that can generate multi-step sales forecasts and support better decision-making for small business users.

#### 1.3.1 Problem Statement

The challenge of this research is to design and develop a machine learning–based system capable of generating accurate six-month sales forecasts for a small-scale spice retail business using historical sales data. The system should not only improve prediction accuracy through models such as XGBoost but also provide interpretable and practical outputs that support inventory planning and scenario-based business decision-making.

### 1.4 Research Motivation

The motivation behind this research arises from the practical challenges faced by small-scale retail businesses in accurately predicting future sales and managing inventory effectively. In niche domains such as spice retailing, demand is highly influenced by seasonal patterns, cultural events, and changing customer behavior, making traditional forecasting methods unreliable. This often leads to inefficient stock management and missed business opportunities. From a technical perspective, the project is motivated by the need to apply advanced machine learning techniques, particularly XGBoost, to address the complexity of multi-step time series forecasting. Predicting sales over a six-month horizon requires capturing non-linear patterns and temporal dependencies within limited historical data, which presents a significant challenge. In addition, there is a need to move beyond pure prediction and provide outputs that are understandable and usable by non-technical users. Therefore, this project is further motivated by the goal of combining accurate forecasting with interpretable insights and simple scenario-based analysis, enabling small business owners to make more informed and practical decisions.

### 1.5 Existing Work

Research in retail sales prediction is extensive, with recent literature shifting heavily towards Machine Learning (ML) and Deep Learning (DL) models. The table below summarizes five highly relevant pieces of existing work, focusing on their contributions to model efficacy, handling complex data, and application in forecasting and inventory planning.

| Citation                   | Summary                                                                                                                                                                       | Limitation                                                                                                                                                                                    | Contribution                                                                                                                                                                                  |
|----------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| (Yadav & Barde 2025)       | Proposes an integrated system using LSTM for time-series forecasting and XGBoost for predictive analytics, featuring real-time data ingestion from POS systems.               | The proposed architecture is highly complex, relying on technologies like Apache Kafka and Hadoop, which may be resource-intensive and impractical for a small business.                      | Demonstrates a powerful hybrid model approach that combines the strengths of different algorithms and integrates real-time capabilities.                                                      |
| (Ganguly & Mukherjee 2024) | Found that an optimized Random Forest model significantly outperformed other methods in handling high seasonality and a large number of product families in a retail setting. | The study focuses primarily on the accuracy of the forecast itself and does not detail the integration of these predictions into subsequent operational planning.                             | Provides strong evidence for the effectiveness of tree-based ensemble models, specifically Random Forest, for complex retail datasets with strong seasonal patterns.                          |
| (Wen et al. 2024)          | Applied an XGBoost model for sales prediction in a Big Mart context, demonstrating superior performance compared to other models.                                             | The research is centered on a large retail chain, and its findings may not be directly transferable to a small, single-store business with different data characteristics.                    | Reinforces the standing of XGBoost as a state-of-the-art model for large-scale retail sales forecasting.                                                                                      |
| (Ecevit et al. 2023)       | Conducted a comparative analysis of LSTM, Prophet, and SARIMA models for short-term (hourly) sales forecasting for a grocery e-commerce retailer.                             | The focus on hourly, e-commerce data may not reflect the patterns of long-term (6-month), in-store sales for a specialty shop.                                                                | Showed that LSTM models can outperform both traditional statistical models (SARIMA) and modern automated models (Prophet) for short-term grocery sales.                                       |
| (Paliari et al. 2021)      | Compared optimized LSTM, XGBoost, and ARIMA models for financial time series (stock index) forecasting.                                                                       | The domain is financial markets, not retail sales. The study also noted that ARIMA performed better on datasets with a limited range of values, a potential consideration for small datasets. | Provides a valuable cross-domain comparison, showing that while ML/DL models are generally superior, traditional statistical methods can still be competitive under specific data conditions. |

Table 1 - Existing works

## 1.6 Research Gap

Despite significant advancements in machine learning–based sales forecasting, several key research gaps persist that justify the need for this project.

First, most existing studies rely on generalized datasets from large retail chains such as Walmart or BigMart, overlooking the specific challenges of specialized retail sectors. There remains limited exploration of niche markets like food ingredients or spices, where sales patterns, product perishability, and seasonal demand introduce unique forecasting complexities (Davinder Pal Singh, 2023).

Secondly, prior research has largely focused on improving predictive accuracy while neglecting the operational and managerial usability of model outputs. Many studies conclude at the evaluation stage (using metrics such as RMSE or MAE), without addressing how such models can be integrated into day-to-day business processes like inventory planning and decision support (Padmanabhan Venkateela, 2025).

In addition, existing approaches rarely incorporate user-adjustable business parameters, limiting the flexibility of models in real-world scenarios. Small business owners often require the ability to explore different “what-if” conditions, such as changes in demand or cost assumptions, which is not sufficiently addressed in current research.

Finally, high-performing algorithms such as XGBoost often suffer from limited interpretability, creating a “black-box” perception that reduces managerial trust and hinders strategic decision-making. A clear need exists for approaches that not only achieve high accuracy but also enhance transparency and explainability in model outcomes (Davinder Pal Singh, 2023).

To address these gaps, this project proposes the development of an interpretable and application-oriented XGBoost-based forecasting system tailored to the spice retail domain. The system integrates SHAP (SHapley Additive Explanations) to visualize feature contributions and incorporates scenario-based analysis to improve practical usability, thereby contributing to both the practical and academic dimensions of predictive analytics.

## 1.7 Contribution to Body of Knowledge

This research will contribute to the body of knowledge in two key areas: the problem domain of small business management and the research domain of applied machine learning.

### 1.7.1 Problem Domain Contribution

This project will deliver a comprehensive and actionable system that empowers small specialty retailers, specifically within the spice retail domain, to leverage their own financial data for advanced forecasting and operational optimization. The system introduces a 6-month multi-step-ahead sales forecasting approach, tailored to capture seasonal and domain-specific sales patterns in niche retail environments.

The outcome is a clear methodology that not only improves inventory management, enhances financial stability, and increases profitability, but also incorporates a scenario-based decision

support system, allowing business users to explore “what-if” conditions and make informed, data-driven decisions. This provides a practical competitive advantage in a challenging market.

### 1.7.2 Research Domain Contribution

This research contributes a detailed comparative analysis of modern machine learning models on a small-business dataset, identifying key features and modeling techniques suitable for long-horizon forecasting in niche retail domains. More significantly, it proposes and validates a practical methodology for 6-month multi-step-ahead forecasting using XGBoost, addressing the challenges of limited and seasonal data.

In addition, the study integrates SHAP (SHapley Additive Explanations) to enhance model interpretability, reducing the “black-box” nature of machine learning models and improving trust for business users. Furthermore, it introduces a structured approach for integrating forecast outputs into financial and inventory management formulas, along with a scenario-based decision framework, thereby bridging the gap between predictive modeling and real-world business application.

## 1.8 Research Challenges

This project will address several inherent research challenges:

1. **Data Scarcity and Quality:** Ensuring the developed models can generate effective forecasts with potentially limited or sparse historical sales data (a common SME constraint) while maintaining high accuracy, as model performance is heavily reliant on data quality.
2. **Computational Resources:** The training and optimization (hyperparameter tuning) of complex ML models necessitate sufficient computational power, which must be addressed, potentially by leveraging cloud platforms or robust local hardware.
3. **Model Selection and Optimization:** Identifying the single best-performing model or optimizing a robust hybrid strategy capable of balancing the capture of linear trends, non-linear patterns, and significant seasonality over the challenging 6-month forecasting horizon.
4. **Integration of Forecasts:** Developing a robust and validated methodology to translate a statistical forecast into dynamic parameters for financial and inventory formulas requires a careful synthesis of data science and business management principles.

## 1.9 Research Questions

RQ1: Which machine learning (ML) or hybrid model (e.g., XGBoost, LSTM, RF, or combined approaches) yields the highest accuracy when performing multi-step-ahead sales forecasting (e.g., a six-month horizon) for seasonal food ingredient retail data?

RQ2: To what extent does the accurate 6-month sales forecast derived from the selected ML model translate into quantifiable improvements in inventory forecasting and cash flow analysis efficiency for the retail business?

RQ3: What are the most effective features for improving the accuracy of a 6-month sales prediction in this domain?

RQ4: How can a multi-step machine learning sales forecast be systematically integrated into standard operational formulas to provide Sales Forecasting, inventory forecasting, and cash flow analysis for a small business?

## 1.10 Research Aim

The primary aim of this research is to design, develop, and evaluate a machine learning–based predictive system for generating accurate six-month sales forecasts for a specialty food retail business. The study focuses on the application of the XGBoost model to capture complex sales patterns and provide reliable predictions. In addition, the system aims to support inventory planning and business decision-making through interpretable outputs and scenario-based analysis.

## 1.11 Research Objectives

| Objectives             | Research Objective (R0x)                                                                                                                                                        | Description                                                                                                                                                                                           | LOs Mapped  | RQ Mapped |
|------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------|-----------|
| Problem Identification | R01: To establish the current limitations of conventional sales forecasting methods for specialty food SMEs and define the requirement for a multi-objective predictive system. | Define the gap regarding accessible ML solutions for Small and Medium Enterprises (SMEs) and the need for translating long-horizon sales forecasts into actionable financial and inventory decisions. | LO1,L<br>O2 | -         |
| Literature Review      | R02: To conduct an in-depth survey comparing advanced ML, DL, and hybrid approaches for retail sales time series forecasting.                                                   | Critically review the documented performance of algorithms such as XGBoost, LSTM, LR and other models for handling seasonality, non-linearity,                                                        | LO1,L<br>O2 | RQ1       |

|                                |                                                                                                                                                                                       |                                                                                                                                                                                                                    |             |             |
|--------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------|-------------|
|                                |                                                                                                                                                                                       | and exogenous variables inherent in retail sales data.                                                                                                                                                             |             |             |
| Requirement Elicitation        | R03: To define the system, data, and functional requirements for integrating the sales prediction output with core business modules (Inventory, Cashflow).                            | Identify necessary transactional data, product financial attributes (like MRP), and temporal features (seasonality) required for modeling and determine the target accuracy level for practical implementation.    | LO2,L<br>O3 | RQ2,<br>RQ3 |
| System Design                  | R04: To design the multi-tiered system architecture and process flow necessary for data preprocessing, optimal ML model selection, and the derivation of actionable business outputs. | Develop a design detailing the required data pipeline, feature engineering steps, model deployment, and the logic flow connecting the sales volume forecast to the calculation of financial and inventory metrics. | LO3         | RQ3         |
| Implementation: Data           | R05: To collect and preprocess sales datasets, focusing specifically on engineering influential features to maximize the model's predictive accuracy.                                 | Prepare the unique food ingredient sales data by handling missing values, incorporating time-based features (lags), and ensuring the integration of available financial details.                                   | LO3,L<br>O4 | RQ2         |
| Implementation: Model          | R06: To implement, tune, and comparatively evaluate multiple advanced ML/DL models to determine the optimal algorithm for accurate 6-month multi-step-ahead sales forecasting.        | Train and optimize competing models (e.g., XGBoost, LSTM, LR) through hyperparameter tuning and cross-validation to select the most robust predictor for the specified long forecasting horizon.                   | LO3,L<br>O4 | RQ1,<br>RQ2 |
| Implementation: Business Logic | R07: To develop the business logic that systematically translates the accurate 6-month sales forecast (R06 output) into specific, quantifiable estimates for Inventory                | Implement the derived logic to calculate inventory requirements, revenues, and subsequent cashflow scenarios and the predicted sales volumes of the 6-month period.                                                | LO3,L<br>O4 | RQ3,<br>RQ4 |

|               |                                                                                                                                                                                                                           |                                                                                                                                                                             |     |     |
|---------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----|-----|
|               | Forecasting and Cashflow analysis.                                                                                                                                                                                        |                                                                                                                                                                             |     |     |
| Testing       | R08: To enhance model interpretability through SHAP-based explanation techniques..                                                                                                                                        | Implement SHAP visualizations to identify key drivers influencing the ML model's sales forecasts, ensuring transparency and actionable insights for business users.         | LO4 | RQ4 |
| Documentation | R09: To document the comprehensive methodology, implementation details, challenges faced, and a synthesis of the research findings.                                                                                       | Compile the final technical report, focusing on the justification for model choice and the practical implications of the ML-based framework in the specialty retail domain. | LO5 | -   |
| Publication   | R10: To prepare a research paper suitable for publication detailing the comparative study of ML/DL models for long-horizon retail sales forecasting and the framework for translating predictions into business outcomes. | Summarize the empirical results comparing ML models for multi-step-ahead forecasting and the integrated business solution for submission to an appropriate academic venue.  | LO5 | -   |

Table 2 - Research Objectives

## 1.12 Project Scope

### 1.12.1 In Scope

- The project will utilize the historical financial dataset from a single food ingredients (spices) shop.
- The core deliverable is a machine learning model capable of generating a six-month sales forecast.
- The forecast will be used as a direct input to generate actionable plans for:
- Sales Forecasting for 6 Months
- Predictive inventory forecasting
- cash flow analysis.
- The research will include a comparative analysis of several machine learning algorithms to identify the most suitable model.

### 1.12.2 Out Scope

- The models will not incorporate broad external macroeconomic indicators (e.g., GDP, inflation rates).
- The analysis is limited to a single-store context and will not be generalized to a multi-store or franchise model.
- The project will not address unpredictable supply-side disruptions beyond incorporating standard lead time variability into inventory calculations.
- The developed system will operate on historical and batch-processed data, not a real-time application integrated or dynamic inventory adjustments.

## 1.13 Hardware/Software Requirements

### 1.13.1 Hardware Requirements

| Component                   | Specification                                                                     |
|-----------------------------|-----------------------------------------------------------------------------------|
| Development Environment CPU | Intel (R) Core (TM) i5 8th generation or higher, such as Intel i9 or AMD Ryzen 9. |
| RAM (Memory)                | 16 GB RAM (Minimum) or 32 GB RAM (Optimal).                                       |
| GPU/VRAM (Optional)         | NVIDIA RTX series (RTX 2080 or newer) or equivalent.                              |
| Storage                     | 1 TB SSD (for high-speed access) or 40 GB minimum disk space.                     |

Table 3 - Hardware Requirement

### 1.13.2 Software Requirements

| Category                   | Software/Tools/Libraries                                                                                                                                               |
|----------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Programming Languages      | <b>Python</b> – primary language for data preprocessing, model development, and evaluation.<br><b>ReactJS</b> -for simple dashboard or result visualization interface. |
| Machine Learning Libraries | <b>XGBoost</b> – main algorithm for regression-based sales forecasting.                                                                                                |
| Development Environment    | <b>Visual Studio Code (VS Code)</b> – for structured coding, version control, and integration with GitHub.                                                             |
| Database/Storage           | <b>CSV-based dataset storage and processing</b>                                                                                                                        |
| ML Frameworks/API          | <b>XGBoost API</b> – for gradient boosting model training and prediction                                                                                               |
| Documentation              | <b>Google Docs</b> – for preparing reports, requirement specifications, and analysis documentation.                                                                    |

Table 4 - SoftwareRequirement

### **1.14 Chapter Summary**

This chapter has made the research project grounded. It started with the determination of the dire necessity of the proper sales forecasting in retail settings and spelled out the precise issue of the implementation of predictive analytics into operational planning. This study is motivated by the fact that it involves technical issues when complex sales data with little historical data is to be modeled. An examination of literature on the subject showed that there was a gap in small business based holistic frameworks, which go beyond prediction and to a practical application. This project is, therefore, focused on developing, evaluating, and designing such a framework based on particular research questions and objectives. The chapter finished with the definition of the exact scope of the project and the list of the technical resources that should be used to complete the project successfully.

## Chapter 2: Literature Review

### 2.1 Chapter Overview

This chapter is the systematic review of the available body of knowledge about sales forecasting, machine learning (ML) methods, and their importance in defining the inventory optimization and financial planning in the retail industry. This review aims at providing the theoretical basis of the intended project, follow the historical development of the forecasting techniques, and critically assess the capabilities and limitations of the models available. Combining these results, the chapter will support the choice of ML methods used when working with complex retail data and reveal the key gaps in research that the combination of 6-month sales predictions with this particular output such as financial planning, resources allocation, inventory forecasting, and cashflow analysis are expected to fill.

### 2.2 Problem Domain

Accurate sales forecasting is a cornerstone of retail operations. In the modern digital age, the conventional approach to business analysis has been transformed by advancements in machine learning and predictive analytics (Ahaggach, Abrouk and Lebon, 2024). ML techniques allow organizations to analyze historical data to identify patterns and trends, generating accurate prognoses relating to future activities. This is particularly critical in competitive markets where profit margins are pressured (Ganguly & Mukherjee, 2024).

#### 2.2.1 The Challenge of Retail Sales as a Time Series Problem

Sales forecasting is fundamentally approached as a time series problem. Time series forecasting consists of predicting several future observations of a given sequence of historical data. In retail environments, sales data are often noisy and non-stationary, exhibiting complex trends, seasonal cycles, and volatilities that change over time (Ahaggach, Abrouk and Lebon, 2024).

Historically, forecasting relied on statistical methods, but these often fall short when dealing with the non-linear relationships and complexity inherent in sales dynamics. For instance, Linear Regression (LR) models often exhibit suboptimal performance, with some studies finding  $R^2$  values around 0.5, indicating that only about half of the variability in sales was explained by the linear model. This inadequacy underscores the necessity for more robust, non-linear models capable of handling intricate sales dynamics (Ganguly & Mukherjee, 2024; Fredén & Larsson, 2020).

Furthermore, the specific objective of predicting sales six months ahead constitutes a multi-step-ahead forecasting problem. This challenge requires careful selection of algorithms and strategies (such as recursive or direct forecasting methods) to manage complexity and maintain accuracy over a long forecast horizon (Souhaib Ben Taie, 2014).

### 2.2.2 Progression from Statistical Models to Machine Learning

The forecasting field has experienced an "explosive increase" in the adoption of machine learning and deep learning techniques since 2013, moving beyond traditional statistical models like ARIMA (Ahaggach, Abrouk and Lebon, 2024). Machine learning algorithms automatically derive rules and decision models from data, allowing them to capture complex patterns using available temporal and historical features, while also supporting the inclusion of external variables where available.(Martins & Galeale, 2023),(Fredén & Larsson, 2020).

Prominent ML and Deep Learning (DL) models repeatedly demonstrated effectiveness in sales prediction across various retail datasets (e.g., BigMart Sales dataset or Walmart sales data from Kaggle):(Davinder Pal Singh , 2023)

- **Ensemble and Tree-based Models:** Models like XGBoost, Random Forest (RF), and Gradient Boosting (GB) are highly favoured. XGBoost, described as a scalable tree boosting system, has proven to be a successful model for forecasting. Studies have shown tuned RF models achieving high performance metrics, such as an  $R^2$  value of 0.945 for sales forecasting. XGBoost has also achieved high accuracy, for example, 84.7% in comparative supermarket sales studies.(Davinder Pal Singh , 2023)
- **Deep Learning (DL) Models:** Long Short-Term Memory (LSTM) neural networks are essential for time series forecasting, especially due to their ability to capture long-term dependencies and sequential patterns. LSTM and XGBoost showed the most promise overall in one comparative study.(Fredén & Larsson, 2020)
- **Hybrid Models:** The synthesis of traditional models with ML algorithms is an emerging trend aimed at enhanced predictive performance and robustness. For example, studies have utilized ensembles that integrate models like LightGBM, XGBoost, and Deep Neural Networks (DNN) to capture both short-term fluctuations and long-term trends.(Tian jin ,2025).

### 2.2.3 Justification for Business-Oriented Outputs

The ultimate value of accurate sales prediction lies in its practical application to managerial decision-making (Padmanabhan Venkateela, 2025). The proposed outputs of this research—sales prediction, inventory forecasting, and decision-support insights—are directly supported as crucial applications in the literature:

1. **Inventory Forecasting:** Improved sales forecasts enable businesses to minimize waste caused by overstocking and maximize sales by preventing understocking (Fredén & Larsson, 2020). For industries dealing with perishable goods such as food ingredients, accurate forecasting is essential to reduce revenue loss. It is particularly important for Small and Medium Enterprises (SMEs) to maintain profitability and achieve long-term growth by reducing risks associated with inventory mismanagement (Y.K. Anuda Harishchandra, 2025).

2. **Business Planning and Resource Allocation:** Sales forecasts act as key inputs for supporting business planning and operational decisions, including resource allocation and demand management. Forecasting models assist managers in making informed strategic decisions by providing future-oriented insights rather than relying on intuition (Ahaggach, Abrouk and Lebon, 2024). In this context, the system supports simple scenario-based analysis, allowing users to explore different business conditions and improve decision-making.

#### 2.2.4 Research Gaps Addressed by the Proposed Project

Despite the proliferation of high-accuracy ML models, the existing research presents specific gaps that this project aims to address:

1. **Gap in Specialised Retail Contexts:** A large proportion of academic research relies on generalised datasets (e.g., BigMart, Walmart sales data). There is a recognized need for more diverse datasets across various market segments to validate the generalizability of predictive models. By focusing on food ingredients (spices), the project addresses the unique sales dynamics and short shelf-life complexities of a specialized retail niche (Davinder Pal Singh, 2023).
2. **Gap in Operationalization and Usability:** Many research efforts focus solely on predictive accuracy metrics (e.g., RMSE, MAE) but do not effectively bridge the gap between technical output and practical business application (Padmanabhan Venkateela, 2025). Research often stops at model evaluation, without addressing the challenges in deployment, integration, and usability for non-technical business stakeholders. This project addresses this gap by providing actionable outputs such as sales forecasting, inventory planning support, and scenario-based decision analysis, thereby improving real-world usability (Padmanabhan Venkateela, 2025).
3. **Gap in Model Interpretability for Decision Making:** High-performing ML models like XGBoost are frequently categorized as non-transparent black-box models (Bohanec, Kljajič Borštnar, & Robnik-Šikonja, 2016). Limited research exists on the interpretability of these models, which is necessary for business environments where managers require transparency and comprehensibility to trust the forecast and justify strategic decisions (Davinder Pal Singh, 2023). Recent advancements in explainable artificial intelligence have introduced techniques such as SHAP (SHapley Additive Explanations), which provide feature-level interpretability for complex models like XGBoost.

## 2.3 Existing Work

The progression of research in sales forecasting has transitioned from relying on simpler statistical models to embracing complex, powerful machine learning (ML) and deep learning (DL) algorithms. This shift has been driven by the need to handle the increasing volume and complexity of transactional retail data. (Davinder Pal Singh, 2023)

### 1. Singh (2023)

This study, focusing on the BigMart Sales dataset, aimed to predict sales patterns and consumer behaviour using machine learning techniques in the business sector. The primary methodology involved a comparative classification study, applying and evaluating models such as XGBoost, K-Nearest Neighbours (KNN), Generalized Linear Model (GLM), and Decision Tree. A comprehensive pipeline was followed, including data preprocessing steps like feature creation, feature reduction via PCA (Principal Component Analysis), and handling missing values. The core contribution was demonstrating the predictive capacity of KNN, with an "upgraded KNN customer categorisation model" achieving an impressive accuracy greater than 0.925. However, the study's limitation lies in its primary focus on *classification* (predicting product categories or behaviour) rather than the *regression* task of predicting precise sales values, which is essential for accurate inventory ordering and financial reporting in your food ingredients shop.

### 2. Ganguly & Mukherjee (2024)

This research addressed the imperative of improving retail sales prediction accuracy by rigorously optimizing various ML models to navigate complex challenges like seasonality and numerous product families. The authors utilized a dataset provided by Favorita Stores, a retail chain in Ecuador. Their methodology centered on ensemble methods and meticulous optimization techniques, comparing models such as Linear Regression (LR), Random Forest (RF), Gradient Boosting (GB), and XGBoost. The core contribution was the achievement of significantly improved results, where an optimized Random Forest (RF) model was able to explain 94.5% of the variance in sales ( $R^2$  of 0.945). The critique highlights the inadequacy of simpler approaches, noting that initial LR application showed overall suboptimal performance, with adjusted  $R^2$  values frequently stabilizing around 0.5. This necessitates the use of complex, computationally intensive methods like optimized RF to achieve real-world viability.

### 3. Ben Taieb (2014)

The primary objective of this doctoral work was to contribute to the development of robust multi-step-ahead forecasting strategies based on machine learning algorithms. This is directly relevant to your 6-month prediction objective. The study systematically compared traditional forecasting approaches, specifically analysing the difference between recursive (iterating a one-step-ahead model) and direct (a separate model for each horizon) strategies. The key contribution involved the introduction of multi-stage forecasting strategies—such as rectify and boost strategies—designed to address the difficult task of choosing between recursive and direct forecasts by combining the best properties of both. The analysis relied on decomposing the Mean Squared Forecast Errors (MSE) into its bias and variance components, a rigorous methodology for evaluating error sources over longer horizons. A persistent limitation noted in the broader literature is the limited attention given to multi-step

forecasts generated specifically with machine learning models, outside of early work on Neural Networks, due to analytical and computational complexity.

#### 4. Ahaggach, Abrouk, and Lebon (2024)

This systematic mapping study (SMS) provided a comprehensive overview of the sales forecasting landscape between 2013 and 2024, meticulously reviewing over 1,000 papers. The research objective was to construct a defined taxonomy of technologies used, scrutinize publication venues, and decode the key factors driving sales forecasting. The core contribution lies in its synthesis of trends, confirming that ML algorithms, statistical methods, and time series analysis are central to modern sales forecasting. The study highlights a remarkable, sustained growth in research efforts, particularly since 2016. In terms of practical data, the review established that time series data are the most frequently used in sales forecasting studies, a natural consequence of sales being inherently time-dependent. The primary limitation of a mapping study is that it provides a descriptive analysis of existing work and does not implement a novel predictive algorithm itself or offer comparative performance metrics of new models.

#### 5. Harishchandra (2025)

This research focused on implementing an Inventory Management System with sales prediction specifically for Small and Medium Enterprises (SMEs). The motivation stemmed from the critical observation that SMEs often rely on rudimentary computational or manual methods, and existing sophisticated forecasting models are often too complex and expensive for them. The methodology proposed the use of XGBoost and features such as Cyclic Encoding to capture seasonal variations. The main contribution is the development of a low-cost, easy-to-use predictive inventory system tailored for SMEs, which focuses on stock balancing to avoid overstocking and understocking. Critically, this research implies that the selection criterion for the model prioritizes feasibility, affordability, and practical utility for small businesses over potentially achieving marginal gains in accuracy from highly complex, resource-intensive models designed for large enterprises.

#### 6. Mustapha and Sithole (2025)

The objective of this research was to examine common machine learning and time series analysis techniques for sales forecasting, ultimately seeking the best-fitted technique for enhancing future profit margins. The methodology involved retrieving five years of publicly available store-item sales data (from Kaggle.com) and training and comparing multiple models, including ARIMA, the deep learning technique LSTM (Long Short-Term Memory) (a recurrent neural network), and XGBoost. The contribution is a comparison of outcomes from these models to establish which is most suitable for time-series forecasting in a retail context. The comparative models employed, such as ARIMA, require that the model be fitted numerous times for each identifier's distinctive value due to the algorithm's adaptability. This highlights a limitation related to computational efficiency and scalability when dealing with a high number of distinct products.

## 2.4 Dataset Selection

The supporting dataset for this project should be comprehensive, primarily focusing on transactional sales data supplemented by financial metrics, product details, and external factors. The complexity of forecasting in the retail sector, particularly for specific categories like food ingredients (spices), requires high-quality data to train predictive models effectively.

A suitable dataset structure, informed by similar supermarket and retail sales forecasting studies, must integrate the following components:

### A. Core Transactional Data (Historical Sales)

The fundamental dataset will comprise the historical sales records, which serve as the primary time series input. This data corresponds directly to the shop's collected financial details of items and is essential for all four projected outcomes (sales prediction, inventory and cashflow).

Example variables include:

Total Monthly Sales Value

Date (Month/Year)

### B. Internal and Engineered Features

To improve model performance, additional features were engineered from the historical sales data to capture temporal patterns and dependencies. These features are essential for handling seasonality and non-linearity within the dataset (Ganguly & Mukherjee, 2024).

To improve model performance, additional features were engineered from the historical sales data to capture temporal patterns and dependencies. These features are essential for handling seasonality and non-linearity within the dataset (Ganguly & Mukherjee, 2024).

Example features include:

- Lagged Sales Data (Lag1, Lag2, Lag3)
- Rolling Statistics (Rolling Mean, Rolling Standard Deviation)
- Time-Based Features (Month Number, Month of Year)

### C. Exogenous Data (External Factors)

Due to data availability constraints, external variables such as weather conditions, promotions, or economic indicators were not included in this study. However, time-related features such as month and seasonality were used to indirectly capture external influences on sales patterns (Martins & Galegale, 2023).

## 2.5 Benchmarking and Evaluation

Evaluating the effectiveness of a sales forecasting model requires comparing its performance against established benchmarks and assessing its reliability using multiple statistical metrics. Selecting the right evaluation metrics is paramount, as different metrics highlight different flaws or benefits in a model.(Fredén & Larsson, 2020)

### 2.5.1 Performance Metrics

For time series regression tasks like sales forecasting, standard metrics are categorized based on whether they prioritize absolute error magnitude or relative percentage error:

| Metric Category                      | Metric                                | Purpose and Significance                                                                                                                                                                                                                                                              |
|--------------------------------------|---------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Magnitude Errors<br>(Absolute Units) | Mean Absolute Error (MAE)             | Represents the average magnitude of error in the units of the sales quantity. It is highly useful in a retail context because it corresponds directly to the actual wasted goods or lost sale opportunities.(Fredén & Larsson, 2020)                                                  |
|                                      | Root Mean Squared Error (RMSE)        | Calculates the square root of the average squared error. RMSE is valuable because it enlarges the effect of larger absolute errors, which are often considered detrimental due to their significant economic effect for a retailer.,(Sithole & Mustapha ,2025)                        |
| Relative/Percentage Errors           | Mean Absolute Percentage Error (MAPE) | Expresses forecast errors as a percentage, which is particularly useful for comparative analysis across products or categories with different sales volumes. XGBoost studies frequently report MAPE reductions as evidence of superior performance.(Ahaggach, Abrouk and Lebon, 2024) |
| Model Fit/Variance                   | $R^2$ -squared ( $R^2$ )              | Represents the proportion of the variance in the target variable (sales) that is explained by the model. High $R^2$ values (e.g., 0.945 achieved by an optimized Random Forest) indicate a strong model fit.(Sithole & Mustapha ,2025)                                                |

Table 5 - Performance Metrics

### 2.5.2 Benchmarking Existing Solutions

Comparison across existing literature confirms that ensemble methods and deep learning techniques generally outperform traditional statistical and linear approaches in handling complex retail sales data:

- **XGBoost and Ensemble Methods:** XGBoost is recurrently identified as a highly effective model for sales forecasting due to its robustness and ability to handle non-linear data. In comparative studies using retail data, XGBoost outperformed models like Decision Tree, Random Forest, Gradient Boosting, and AdaBoost, achieving an  $R^2$  of 0.852. In specialized domains like pharmaceutical sales, XGBoost demonstrated significant error reduction, lowering the Mean Absolute Percentage Error (MAPE) from 27.48% to 17.89% for certain product categories compared to Seasonal Naïve models, (Padmanabhan Venkiteela, 2025)
- **Deep Learning (LSTM):** Long Short-Term Memory (LSTM) models are widely applied to time series problems for their capability to capture temporal dependencies. LSTM often shows high performance in terms of accuracy and  $R^2$  values compared to linear models. However, its performance remains context-dependent; in some datasets (e.g., stable prescription trends), simpler models like SARIMA proved superior. (Vili Ripatti, 2024), (Dr. K K Ramachandran, 2023)
- **Linear and Basic Models:** Traditional methods like Linear Regression (LR) and simple Decision Trees often exhibit suboptimal performance, struggling to capture the complex, non-linear dynamics of retail sales, sometimes yielding low  $R^2$  values around 0.5.
- **Random Forest:** Random Forest is another widely used ensemble learning method that performs well in retail sales forecasting due to its ability to reduce overfitting and handle non-linear relationships. Studies have shown that optimized Random Forest models can achieve high predictive performance, especially in datasets with strong seasonality and multiple influencing factors. However, compared to boosting methods like XGBoost, Random Forest may be less effective in capturing complex residual patterns.

In addition to traditional performance metrics, this research also emphasizes the importance of model interpretability in evaluating forecasting effectiveness. To ensure transparency and managerial trust, the project integrates SHAP (SHapley Additive Explanations) as part of the evaluation process.

## 2.6 Chapter Summary

This review has affirmed that the complex retail sales forecasting should preferably be performed on the basis of the most advanced machine learning models, e.g. XGBoost, rather than the conventional methods. The project continues this and puts emphasis on the specialized food ingredients-spices-marketplace and the unique dynamics in the same e.g. short shelf-life. A major goal of this study is to fill the gap between technical prediction accuracy and business intelligence that can be taken to action. The sales forecast will be eventually converted into the outputs required in the financial planning, inventory management and cash-flow analysis. Further work will also be focused on the interpretation of these elaborate models, which will increase trust and eventually its implementation by users in the business fraternity.

## Chapter 03: Methodology

### 3.1 Chapter Overview

This chapter would provide the methodological framework of this research project. This chapter begins by defining the research methodology based on the Research Onion by Saunders and the reason why the philosophical, strategic, and tactical alternatives have been chosen. It then follows it up with discussions in finer details of the development methodology followed- requirement elicitation method, system design, programming, and testing. This chapter also defines the target solution methodology and gives an overview of the machine learning pipeline stepwise starting with the data collection and ending with model evaluation. The section also provides the project management methodology that has been used in driving the project implementation including scope, timeline, resource planning and monitoring/mitigation plans of the risks involved in the project.

### 3.2 Research Methodology

The research methodology chosen for this project follows the multi-layered structure proposed by Saunders, leveraging a combination of quantitative ML analysis and qualitative application assessment, which is specifically suited for an implementation project within a specialized business context.

| Layer of Research Onion | Description in Context of Sales Forecasting Project | Justification and Citation                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
|-------------------------|-----------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Research Philosophy     | Pragmatism                                          | The rationale behind the adoption of this philosophy is that the study incorporates both a quantitative analysis (ML model accuracy) and a qualitative analysis (feedback on the usefulness of generated financial and inventory outputs provided by the stakeholders). Pragmatism has the benefit of using various approaches best suited to address the business issue at hand on a real-world, which is optimizing inventory (Small-Medium Enterprise(SME)).(Y.K.AnudaHarishchandra, 2025) |

|                       |                                             |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
|-----------------------|---------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Research Approach     | Deductive Approach                          | This research aims to assess and confirm the efficiency of some of the well-known forecasting models, such as XGBoost or LSTM, on a special proprietary dataset. We do not want to develop new theoretical frameworks, but rather determine whether these current machine learning models can help us make the right predictions of sales, to the level of high accuracy we expect. (Niluja , 2023)                                                                                                                                      |
| Methodological Choice | Mixed Method (Quantitative and Qualitative) | This paper aims at testing and verifying the applicability of certain popular forecasting models, such as XGBoost and LSTM, on a particular proprietary dataset. There is no effort to construct new theoretical concepts; the question is whether these already existing machine learning models are capable of rising to the challenge of making highly accurate sales predictions or not. In essence, we would like to know whether these models that have been established will bring the results we are expecting..(Niluja , 2023)  |
| Research Strategy     | Experimentation and Case Study              | We will measure our project with two major approaches. To measure the performance of our machine learning models, first, quantitative techniques will be used, such as experimentation, to evaluate the performance using such measures as MAE and RMSE. Second, we will obtain qualitative feedback by interviewing to determine the usability and feasibility of our outputs inventory forecasting and cash flow to non technical parties. This makes it possible to have effective solutions that are easy to use. (Fourkiotis ,2024) |

|                                             |                                                       |                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
|---------------------------------------------|-------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Time Horizon                                | Longitudinal Study                                    | We will assess our machine learning models using a combination of quantitative and qualitative approaches. To measure technical performance, we'll evaluate the Mean Absolute Error (MAE) and Root Mean Square Error (RMSE) through experiments. To ensure the outputs we generate—Financial Planning and Inventory Forecasting—are friendly for non-technical end users, we'll conduct non-technical feedback interviews.(Souhaib Ben Taieb, 2014)     |
| Techniques and Procedures (Data & Analysis) | Machine Learning (ML) Pipeline and Data Visualization | The process follows standard machine learning procedures consisting of data pre-processing (including missing value treatment and time-series component feature engineering), model selection and training (testing algorithms such as XGBoost and LSTM) and evaluation (using RMSE and MAPE). Visualizing data through graphics like heat maps and scatter plots is required as part of exploratory analysis and for making the final user actionable. |

Table 6 - Research Methodology

### 3.3 Development Methodology

#### 3.3.1 Requirement Elicitation Methodology

The process of identifying and refining the system requirements of the SalesSense project was done as a synthesis of requirement elicitation techniques. The surveys have been used to elicit the input of potential users like the owners of small businesses and the retail employees over usability and features needed. The analysis of the documents was conducted based on the current research papers and other similar systems in order to reveal gaps and functionalities. System features were refined in brainstorming sessions and brought in line with project objectives. Also, it was observed that retail operations would contribute to the comprehension of real-life issues associated with tracking sales and inventory management.

Functionality of the system was also visualized by prototyping and feedback given to complete the improvement.

### 3.3.2 Design Methodology

SSADM (Structured Systems Analysis and Design Methodology) or OOADM (Object-Oriented Analysis and Design Methodology) depending on system structure.

### 3.3.3 Programming Paradigm

A combination of programming paradigms was used in the development of the system to ensure flexibility and efficiency. The implementation primarily follows Object-Oriented Programming (OOP) principles, enabling modular design and code reusability in the backend developed using Python. Structured programming was applied for handling data preprocessing and model execution tasks, ensuring a clear and logical flow of operations. In addition, functional programming concepts were utilized in data transformation and feature engineering processes, particularly for handling time series data. The frontend of the system was developed using React, following a component-based architecture to support modular and maintainable user interface design.

### 3.3.4 Testing Methodology

System test methodology of the system is in form of both model-level and system-level testing. A model testing was done to understand the accurateness of the forecasting model using the regression measures of RMSE, MAE, MAPE, and  $R^2$ . Also, the prototype testing was conducted to evaluate the work of the whole system, API functionality, data processing, and interaction with the interface. This hybrid solution guarantees technical accuracy and healthy functionality of the system.

### 3.3.5 Solution Methodology

1. Dataset Collection: historical monthly sales dataset collected from the retail shop
2. Data Preprocessing: Clean and preprocess data, ensuring it is ready for training.
3. Feature Selection and Engineering: Extract time-based and lag features such as previous sales (Lag1, Lag2, Lag3), rolling statistics, and seasonal indicators
4. Model Selection: Select an appropriate model (e.g., XGBoost) based on performance metrics such as RMSE and MAE.
5. Model Training: Train the model using the preprocessed dataset.
6. Testing: Evaluate the model's performance metrics and interpretability analysis will be conducted using the SHAP framework.
7. Feedback Loop: Refine the model based on evaluation results and performance metrics.

### 3.4 Project Management Methodology

Given the limitations of a final year timeline, the project management methodology chosen for this research project is essential for handling the intricate, iterative nature of Machine Learning (ML) development. The strategy is usually based on the Agile methodology, which is popular due to its high level of stakeholder interaction, iterative process, and adaptability to changing output requirements as the model is being refined. To strike a balance between the rigorous documentation requirements of academic research and the development flexibility required for an ML model, a hybrid approach, like Agile PRINCE2, can be used.

#### 3.4.2 Schedule

##### 3.4.2.1 Gantt Chart

[Time Schedule](#) displays the visual representation of the Gantt chart.

##### 3.4.2.2 Deliverables: Literature review, prototype development, final report submission.

| Deliverables                                                           | Dates                            |
|------------------------------------------------------------------------|----------------------------------|
| Formative Submission of Initial Project Proposal                       | 10th October 2025                |
| Formative Submission of Literature Review                              | 10th October 2025                |
| Summative Submission of Project Proposal and Requirement Specification | 13th November 2025               |
| Summative Submission of Interim Progress Demonstration                 | 30th January 2026                |
| Summative Submission of Final Project Thesis                           | 30th March 2026                  |
| Final Project Viva                                                     | 26th April 2026 to 20th May 2026 |

Table 7 - Deliverables

### 3.4.3 Resource Requirements

#### 3.4.3.1 Data

The project will utilize historical sales transaction data collected from a local food ingredients (spices) retailer.

This dataset will include:

- Daily and monthly sales quantities, product names, and pricing details (MRP, cost)..

If required, supplementary or synthetic datasets may be used to simulate missing data and enhance model training.

This data will be stored and processed in CSV/Excel format within the code file.

#### 3.4.3.2 Skills

To successfully develop and implement the ML-based forecasting framework, the following technical and analytical skills are required:

- **Machine Learning & Deep Learning** – for developing and comparing models such as XGBoost .
- **Python Programming** – for data preprocessing, feature engineering, model training, and evaluation.
- **Data Science & Statistical Analysis** – for exploratory data analysis (EDA), time series decomposition, and accuracy measurement (RMSE, MAE, MAPE).
- **Data Visualization** – using Matplotlib, Seaborn, or Plotly to create dashboards and graphs for interpretability.
- **Model Explainability** – using SHAP or feature importance to ensure transparency for non-technical stakeholders.

### 3.4.4 Risk and Mitigation

| Risk                                                       | Severity | Frequency | Mitigation Strategy                                                                                                                              |
|------------------------------------------------------------|----------|-----------|--------------------------------------------------------------------------------------------------------------------------------------------------|
| Insufficient training data                                 | 5        | 3         | Use data augmentation or synthetic data generation. Collect additional sales records and external features (e.g., seasonal or promotional data). |
| Poor model accuracy                                        | 4        | 3         | Perform hyperparameter tuning, cross-validation, and comparative testing between XGBoost, and other models.                                      |
| Overfitting                                                | 5        | 4         | Regularization (L1/L2), early stopping, and cross-validation for generalization.                                                                 |
| Hardware limitations                                       | 3        | 2         | Utilize cloud-based environments such as Google Colab Pro or AWS for large-scale training.                                                       |
| Data quality issues (missing, noisy, or inconsistent data) | 4        | 4         | Implement preprocessing pipelines for data cleaning, imputation, and normalization before model training.                                        |
| Stakeholder dissatisfaction                                | 3        | 3         | Periodic feedback sessions and user testing with small business owners and managers for dashboard/report refinements.                            |
| Delays in model training                                   | 3        | 3         | Code optimization, efficient libraries (NumPy, Pandas), and parallel computation.                                                                |
| Model interpretability concerns (black-box perception)     | 3        | 3         | Integrate SHAP or feature-importance visualization for enhanced transparency for business users.                                                 |

Table 8 - Risk &amp; Mitigation

### 3.5 Chapter Summary

The chapter gives a general introduction to the approach that has been employed in the creation of the Machine Learning-based Sales Forecasting System. It adheres to the Research Onion framework by Saunders by utilizing a practical philosophy and a mixed-method approach, which involves the assessment of the models quantitatively and the stakeholders qualitatively. It is possible to divide the development process into requirement elicitation, design, programming, testing, and an ML pipeline between data collection and evaluation. A hybrid Agile-PRINCE2 methodology is applied to project management as it has clear scope, schedule, and resources and plans related to the mitigation of risk to achieve a successful project implementation and dependable forecasting results.

## Chapter 04: System Requirement Specification (SRS)

### 4.1 Chapter Overview

This chapter gives a general overview of the general methodology that has been used in coming up with the Machine Learning-based Sales Forecasting System. It also conforms to the Research Onion framework by Saunders in that it embraces a pragmatic philosophy and mixed-method approach; i.e. evaluating a quantitative model and a qualitative stakeholder response. It can be divided into the development process of requirement elicitation, design, programming, testing, and an ML pipeline of data collection to evaluation. The hybrid Agile-PRINCE2 approach is applied to project management, which specifies the scope, timeline, and resources and plans on how risk can be addressed to improve the execution of projects and ensure successful forecasting results.

### 4.2 Rich Picture Diagram

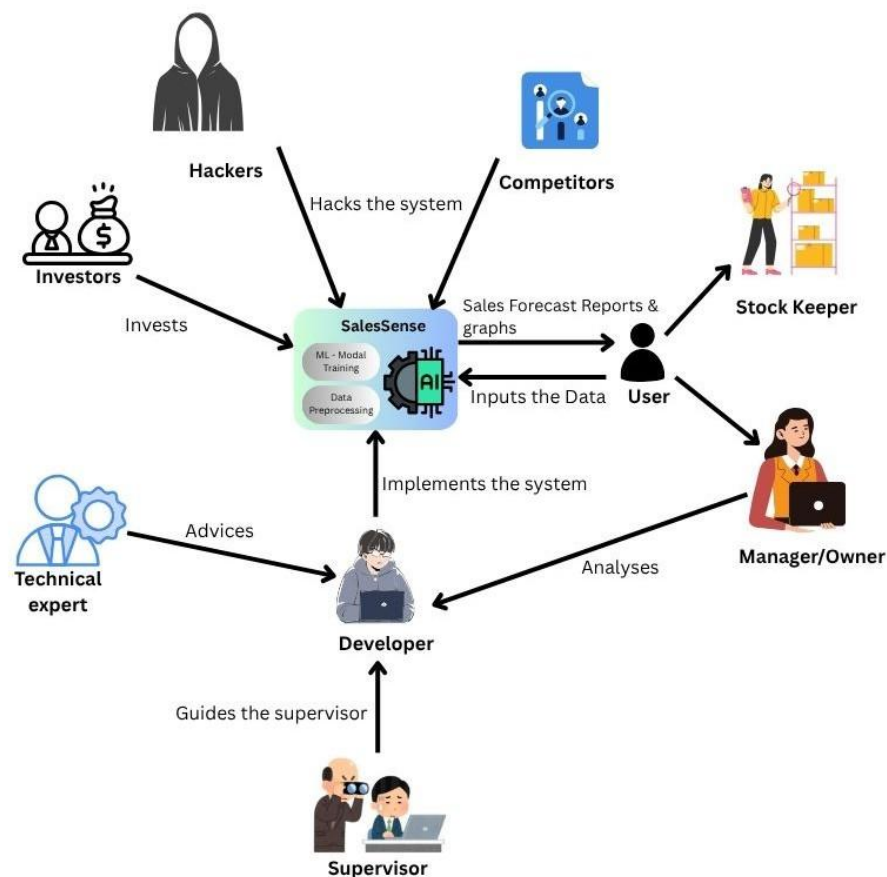


Figure 1 - Rich Picture Diagram

### 4.3 Stakeholder Analysis

#### 4.3.1 Stakeholder Onion Model

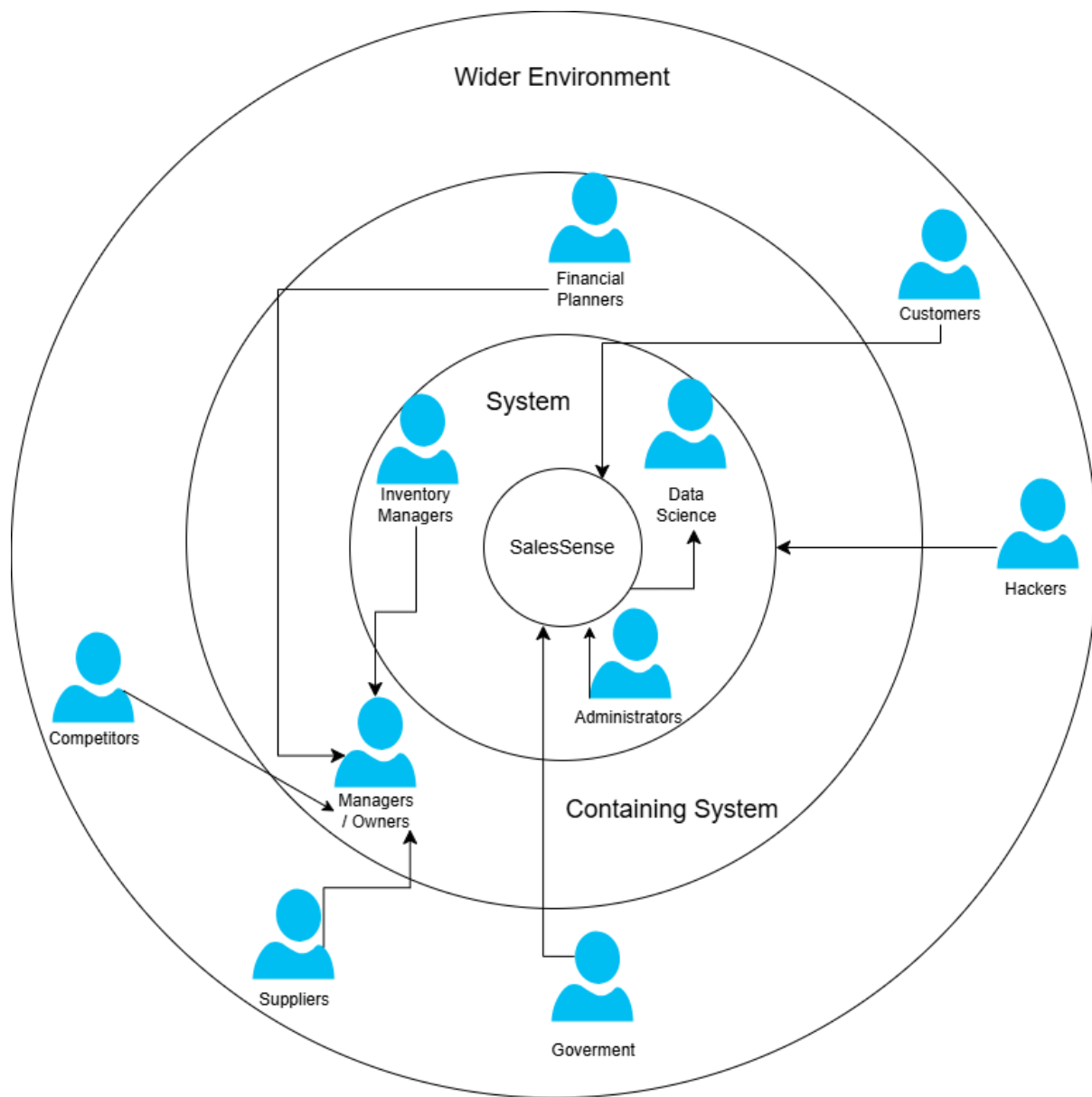


Figure 2 - Onion Modal Diagram

### 4.3.2 Stakeholder Viewpoints

| Stakeholder                        | Role/Category                            | Viewpoint/Expectation                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
|------------------------------------|------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Business Owners / Managers         | Functional Beneficiary / Decision Makers | The strategic planning, operational optimization, and general profitability require accurate sales projections. There should be dependable long-term predictions (a six-month prediction) to help business develop. They must expect that the system will help them make data-driven decisions. Projections are needed so as to enable effective allocation of resources. Need a system reasonably priced and reasonably easy to use? The sales projections are likely to be in a graphical format in order to facilitate in understanding and making decisions. Strive to achieve high prediction accuracy, occasionally at 70 percent or more. |
| Inventory Managers / Stock Keepers | Operational Beneficiary                  | Accurate predictions are needed in order to organize production scheduling efficiently and ensure that inventory is strictly controlled. There should be predictions to prevent shortages and over inventory and this will reduce loss of revenue. Minimize stock-out (waste) and maximize sales (do not understock). The system should schedule items, and it should have an easy-to-use and intuitive user interface (UI) to help a user analyze the sales trends without relying on a complex technical background.                                                                                                                           |

|                                    |                                     |                                                                                                                                                                                                                                                                                                                                                                                                                              |
|------------------------------------|-------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Data Scientists / ML Engineers     | Operational Maintenance / Developer | Focus on machine learning model and code implementation and execution of the system. Algorithms (XGBoost) should be selected and proper training and evaluation done on the model. Successful model training requires the data to be clean and well-formatted (i.e. CSV format). Later addition to make the system better and easier to debug, expect the underlying system design to be modular, scalable and maintainable. |
| Financial Planners / Investors     | Financial Beneficiary               | Look for system output that can help you with financial planning and forecasting into the future. fixed on extracting as much income as possible and delivering a measurable return on investment (ROI).                                                                                                                                                                                                                     |
| Customers / End-Users              | Positive Stakeholder                | Expect greater availability of products to reduce the chance of encountering out-of-stock items. for their demands, needs, and wants to be regularly met in order to ensure they're happy with the provided services.                                                                                                                                                                                                        |
| System Administrators / IT Support | Operational Administration          | Require that the system is reliable, secure and can be easily installed in the work place. To be able to deal with growth in data and user base, it is required there to be a certain level of scalability in the application.                                                                                                                                                                                               |

|                                     |                                    |                                                                                                                                                                                                                                                                                                                                             |
|-------------------------------------|------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Competitors                         | Negative Stakeholder               | With reference to the system features, try to develop a comparable or superior solution (e.g., predicting sales using ML). If the SME has problems with inventory control or the supply chain, they might try to capitalize.                                                                                                                |
| Regulatory Authorities / Government | Political Beneficiary / Regulators | Careful about ensuring the company adheres to the laws and regulations, including taxes, must ensure that the manner in which the company manages its sales and customer data is not violating any of the relevant data privacy and data protection laws (e.g., the GDPR compliance, addressing ethical concerns related to data analysis). |

Table 9 - Stakeholder Viewpoint

## 4.4 Selection of Requirement Elicitation Methodologies

Various methods were used to collect the requirements of the Sales prediction and Inventory Forecasting System based on machine learning in this project. Each methodology was chosen because of the capacity to collect the right, relevant, and viable insights of the stakeholders and experts in the field.

### 4.4.1 Literature review

The Literature Review methodology provided significant information about the body of existing literature and other solutions that the researchers have recently suggested and implemented. This also enabled to get a clearer insight into the reinforcement learning and methods and the research gaps.

### 4.4.2 Surveys / Questionnaires:

Speaking of collecting quantitative data about the user expectations, the features that they would desire in the system, and the visualization forms, surveys were distributed among the potential end users, such as sales analysts and employees. The focus on the dashboard design, frequency of forecasts and notification methods were made in part due to the feedback.

#### 4.4.3 Brainstorming Sessions and Focus Group Discussions:

In order to enhance the system requirement, managers, IT personnel, and small business owners were involved in joint sessions. These discussions generated key ideas in such areas as dashboard customization, user-based access, and integration with external databases.

### 4.5 Discussion of Findings

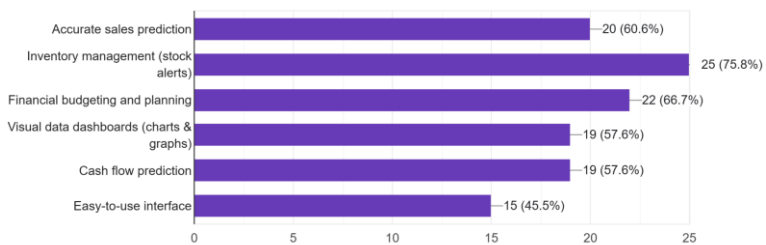
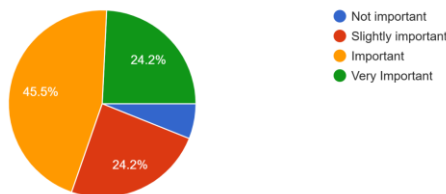
#### 4.5.1 Literature Review Findings

| Citation                                                                        | Findings                                                                                                                                                                                                                                  |
|---------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| (Bohanec, Kljajič Borštnar, & Robnik-Šikonja, 2016), (Davinder Pal Singh, 2023) | Limited research exists on the interpretability of high-performing ML models like XGBoost (often categorized as non-transparent black-box models), which is necessary for managers to trust the forecast and justify strategic decisions. |
| (Davinder Pal Singh, 2023)                                                      | There is a recognized need for more diverse datasets across various market segments to validate the generalizability of predictive models, as much research relies on generalised datasets.                                               |
| (Y.K.Anuda Harishchandra, 2025)                                                 | Accurate forecasting is crucial for Small and Medium Enterprises (SMEs) to maintain profitability and attain long-term business growth by mitigating operational and financial risks associated with inventory mismanagement.             |

Table 10 - Literature Review Findings

## 4.5.2 Survey Findings

| <b>Question</b>                  | <b>How familiar are you with the concept of sales forecasting and machine learning applications in retail business?</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |                   |            |              |       |                   |       |                     |       |               |       |
|----------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------|------------|--------------|-------|-------------------|-------|---------------------|-------|---------------|-------|
| <b>Aim of the Question</b>       | To determine how familiar the respondents are with the concept of sales forecasting and the use of ML in retail.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |                   |            |              |       |                   |       |                     |       |               |       |
| <b>Findings &amp; Conclusion</b> | <p>The survey results indicate that 42.4% of respondents are somewhat familiar, 33.3% are not familiar, and 24.2% are moderately familiar. This shows that a considerable portion of respondents lack a deep understanding of these technologies, validating the need for awareness and simplified tools in this area.</p> <p>How familiar are you with the concept of sales forecasting and machine learning applications in retail business?<br/>33 responses</p> <table border="1"> <thead> <tr> <th>Familiarity Level</th> <th>Percentage</th> </tr> </thead> <tbody> <tr> <td>Not familiar</td> <td>33.3%</td> </tr> <tr> <td>Somewhat familiar</td> <td>42.4%</td> </tr> <tr> <td>Moderately familiar</td> <td>24.2%</td> </tr> <tr> <td>Very familiar</td> <td>0%</td> </tr> </tbody> </table> | Familiarity Level | Percentage | Not familiar | 33.3% | Somewhat familiar | 42.4% | Moderately familiar | 24.2% | Very familiar | 0%    |
| Familiarity Level                | Percentage                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |                   |            |              |       |                   |       |                     |       |               |       |
| Not familiar                     | 33.3%                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |                   |            |              |       |                   |       |                     |       |               |       |
| Somewhat familiar                | 42.4%                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |                   |            |              |       |                   |       |                     |       |               |       |
| Moderately familiar              | 24.2%                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |                   |            |              |       |                   |       |                     |       |               |       |
| Very familiar                    | 0%                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |                   |            |              |       |                   |       |                     |       |               |       |
| <b>Question</b>                  | <b>How useful do you think a machine learning–based forecasting system would be for helping retailers (e.g., spice shops) manage inventory and finances?</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |                   |            |              |       |                   |       |                     |       |               |       |
| <b>Aim of the Question</b>       | To assess how respondents perceive the usefulness of ML-based systems in retail operations.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |                   |            |              |       |                   |       |                     |       |               |       |
| <b>Findings &amp; Conclusion</b> | <p>Findings reveal that 45.5% consider it very useful, 39.4% useful, and 15.2% slightly useful. This indicates strong positive feedback, suggesting that such systems would be highly valued by retailers.</p> <p>How useful do you think a machine learning–based forecasting system would be for helping retailers (e.g., spice shops) manage inventory and finances?<br/>33 responses</p> <table border="1"> <thead> <tr> <th>Usefulness Level</th> <th>Percentage</th> </tr> </thead> <tbody> <tr> <td>Not useful</td> <td>0%</td> </tr> <tr> <td>Slightly useful</td> <td>15.2%</td> </tr> <tr> <td>Useful</td> <td>39.4%</td> </tr> <tr> <td>Very useful</td> <td>45.5%</td> </tr> </tbody> </table>                                                                                            | Usefulness Level  | Percentage | Not useful   | 0%    | Slightly useful   | 15.2% | Useful              | 39.4% | Very useful   | 45.5% |
| Usefulness Level                 | Percentage                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |                   |            |              |       |                   |       |                     |       |               |       |
| Not useful                       | 0%                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |                   |            |              |       |                   |       |                     |       |               |       |
| Slightly useful                  | 15.2%                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |                   |            |              |       |                   |       |                     |       |               |       |
| Useful                           | 39.4%                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |                   |            |              |       |                   |       |                     |       |               |       |
| Very useful                      | 45.5%                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |                   |            |              |       |                   |       |                     |       |               |       |

| Question                                 | Which of the following features do you believe are most important in such a system?                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |                  |            |               |                           |                    |       |                                     |       |                |                                  |    |       |                                          |    |       |                      |    |       |                       |    |       |
|------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------|------------|---------------|---------------------------|--------------------|-------|-------------------------------------|-------|----------------|----------------------------------|----|-------|------------------------------------------|----|-------|----------------------|----|-------|-----------------------|----|-------|
| Aim of the Question                      | To identify which system features respondents prioritize most.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |                  |            |               |                           |                    |       |                                     |       |                |                                  |    |       |                                          |    |       |                      |    |       |                       |    |       |
| Findings & Conclusion                    | <p>Responses were diverse, with many participants selecting combinations including inventory management, accurate sales prediction, financial budgeting, and visual dashboards. The highest combined preference (12.1%) included all major features, implying that users prefer a comprehensive, all-in-one solution.</p> <div><p>Which of the following features do you believe are most important in such a system?<br/>33 responses</p><table><thead><tr><th>Feature</th><th>Count</th><th>Percentage</th></tr></thead><tbody><tr><td>Accurate sales prediction</td><td>20</td><td>60.6%</td></tr><tr><td>Inventory management (stock alerts)</td><td>25</td><td>75.8%</td></tr><tr><td>Financial budgeting and planning</td><td>22</td><td>66.7%</td></tr><tr><td>Visual data dashboards (charts &amp; graphs)</td><td>19</td><td>57.6%</td></tr><tr><td>Cash flow prediction</td><td>19</td><td>57.6%</td></tr><tr><td>Easy-to-use interface</td><td>15</td><td>45.5%</td></tr></tbody></table></div> | Feature          | Count      | Percentage    | Accurate sales prediction | 20                 | 60.6% | Inventory management (stock alerts) | 25    | 75.8%          | Financial budgeting and planning | 22 | 66.7% | Visual data dashboards (charts & graphs) | 19 | 57.6% | Cash flow prediction | 19 | 57.6% | Easy-to-use interface | 15 | 45.5% |
| Feature                                  | Count                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | Percentage       |            |               |                           |                    |       |                                     |       |                |                                  |    |       |                                          |    |       |                      |    |       |                       |    |       |
| Accurate sales prediction                | 20                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | 60.6%            |            |               |                           |                    |       |                                     |       |                |                                  |    |       |                                          |    |       |                      |    |       |                       |    |       |
| Inventory management (stock alerts)      | 25                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | 75.8%            |            |               |                           |                    |       |                                     |       |                |                                  |    |       |                                          |    |       |                      |    |       |                       |    |       |
| Financial budgeting and planning         | 22                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | 66.7%            |            |               |                           |                    |       |                                     |       |                |                                  |    |       |                                          |    |       |                      |    |       |                       |    |       |
| Visual data dashboards (charts & graphs) | 19                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | 57.6%            |            |               |                           |                    |       |                                     |       |                |                                  |    |       |                                          |    |       |                      |    |       |                       |    |       |
| Cash flow prediction                     | 19                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | 57.6%            |            |               |                           |                    |       |                                     |       |                |                                  |    |       |                                          |    |       |                      |    |       |                       |    |       |
| Easy-to-use interface                    | 15                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | 45.5%            |            |               |                           |                    |       |                                     |       |                |                                  |    |       |                                          |    |       |                      |    |       |                       |    |       |
| Question                                 | How important is it for you to understand how the system makes predictions (model transparency)?                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |                  |            |               |                           |                    |       |                                     |       |                |                                  |    |       |                                          |    |       |                      |    |       |                       |    |       |
| Aim of the Question                      | To determine how much users value model transparency and interpretability.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |                  |            |               |                           |                    |       |                                     |       |                |                                  |    |       |                                          |    |       |                      |    |       |                       |    |       |
| Findings & Conclusion                    | <p>About 45.5% of respondents rated it important, 24.2% slightly important, and 24.2% very important, while 6.1% found it not important. This suggests that most users appreciate having insights into how predictions are made, though full transparency is not a top priority for everyone.</p> <div><p>How important is it for you to understand how the system makes predictions (model transparency)?<br/>33 responses</p><table><thead><tr><th>Importance Level</th><th>Percentage</th></tr></thead><tbody><tr><td>Not important</td><td>6.1%</td></tr><tr><td>Slightly important</td><td>24.2%</td></tr><tr><td>Important</td><td>45.5%</td></tr><tr><td>Very Important</td><td>24.2%</td></tr></tbody></table></div>                                                                                                                                                                                                                                                                              | Importance Level | Percentage | Not important | 6.1%                      | Slightly important | 24.2% | Important                           | 45.5% | Very Important | 24.2%                            |    |       |                                          |    |       |                      |    |       |                       |    |       |
| Importance Level                         | Percentage                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |                  |            |               |                           |                    |       |                                     |       |                |                                  |    |       |                                          |    |       |                      |    |       |                       |    |       |
| Not important                            | 6.1%                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |                  |            |               |                           |                    |       |                                     |       |                |                                  |    |       |                                          |    |       |                      |    |       |                       |    |       |
| Slightly important                       | 24.2%                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |                  |            |               |                           |                    |       |                                     |       |                |                                  |    |       |                                          |    |       |                      |    |       |                       |    |       |
| Important                                | 45.5%                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |                  |            |               |                           |                    |       |                                     |       |                |                                  |    |       |                                          |    |       |                      |    |       |                       |    |       |
| Very Important                           | 24.2%                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |                  |            |               |                           |                    |       |                                     |       |                |                                  |    |       |                                          |    |       |                      |    |       |                       |    |       |
| Question                                 | If such a system were available for small retailers, how likely do you think they would adopt it?                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |                  |            |               |                           |                    |       |                                     |       |                |                                  |    |       |                                          |    |       |                      |    |       |                       |    |       |

| <b>Aim of the Question</b>       | To gauge the likelihood of adoption among small retailers.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |          |            |               |      |                    |       |           |       |                |       |
|----------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------|------------|---------------|------|--------------------|-------|-----------|-------|----------------|-------|
| <b>Findings &amp; Conclusion</b> | <p>Results show 57.6% think retailers are likely to adopt the system, 24.2% unlikely, and 18.2% very likely. This suggests a strong market potential with some skepticism possibly due to cost or complexity concerns.</p> <p>If such a system were available for small retailers, how likely do you think they would adopt it?<br/>33 responses</p> <table border="1"> <thead> <tr> <th>Category</th> <th>Percentage</th> </tr> </thead> <tbody> <tr> <td>Very unlikely</td> <td>0%</td> </tr> <tr> <td>Unlikely</td> <td>24.2%</td> </tr> <tr> <td>likely</td> <td>57.6%</td> </tr> <tr> <td>Very likely</td> <td>18.2%</td> </tr> </tbody> </table>                                                 | Category | Percentage | Very unlikely | 0%   | Unlikely           | 24.2% | likely    | 57.6% | Very likely    | 18.2% |
| Category                         | Percentage                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |          |            |               |      |                    |       |           |       |                |       |
| Very unlikely                    | 0%                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |          |            |               |      |                    |       |           |       |                |       |
| Unlikely                         | 24.2%                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |          |            |               |      |                    |       |           |       |                |       |
| likely                           | 57.6%                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |          |            |               |      |                    |       |           |       |                |       |
| Very likely                      | 18.2%                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |          |            |               |      |                    |       |           |       |                |       |
| <b>Question</b>                  | <b>How confident are you in interpreting charts or graphs showing sales trends and forecast results?</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |          |            |               |      |                    |       |           |       |                |       |
| <b>Aim of the Question</b>       | To understand users' confidence in reading visual data representations.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |          |            |               |      |                    |       |           |       |                |       |
| <b>Findings &amp; Conclusion</b> | <p>The findings indicate 42.4% are confident, 30.3% somewhat confident, 21.2% very confident, and 6.1% not confident. This highlights that most users are comfortable with visual insights, supporting the inclusion of data visualization features.</p> <p>How confident are you in interpreting charts or graphs showing sales trends and forecast results?<br/>33 responses</p> <table border="1"> <thead> <tr> <th>Category</th> <th>Percentage</th> </tr> </thead> <tbody> <tr> <td>Not Confident</td> <td>6.1%</td> </tr> <tr> <td>Somewhat Confident</td> <td>30.3%</td> </tr> <tr> <td>Confident</td> <td>42.4%</td> </tr> <tr> <td>Very Confident</td> <td>21.2%</td> </tr> </tbody> </table> | Category | Percentage | Not Confident | 6.1% | Somewhat Confident | 30.3% | Confident | 42.4% | Very Confident | 21.2% |
| Category                         | Percentage                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |          |            |               |      |                    |       |           |       |                |       |
| Not Confident                    | 6.1%                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |          |            |               |      |                    |       |           |       |                |       |
| Somewhat Confident               | 30.3%                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |          |            |               |      |                    |       |           |       |                |       |
| Confident                        | 42.4%                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |          |            |               |      |                    |       |           |       |                |       |
| Very Confident                   | 21.2%                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |          |            |               |      |                    |       |           |       |                |       |
| <b>Question</b>                  | <b>How important do you think visualization features (such as trend graphs, prediction comparison charts, or dashboards) are in understanding the system's output?</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |          |            |               |      |                    |       |           |       |                |       |
| <b>Aim of the Question</b>       | To evaluate the perceived importance of visualization in the system.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |          |            |               |      |                    |       |           |       |                |       |

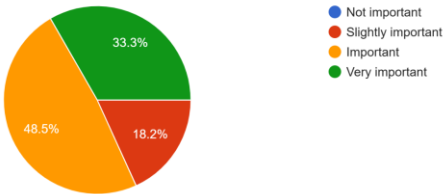
| <b>Findings &amp; Conclusion</b> | <p>The majority (48.5%) rated visualization as important, 33.3% as very important, and 18.2% as slightly important. The results emphasize that visual outputs are essential for user understanding and engagement.</p> <p>How important do you think visualization features (such as trend graphs, prediction comparison charts, or dashboards) are in understanding the system's output?<br/>33 responses</p>  <table border="1"><thead><tr><th>Importance Level</th><th>Percentage</th></tr></thead><tbody><tr><td>Not important</td><td>0%</td></tr><tr><td>Slightly important</td><td>18.2%</td></tr><tr><td>Important</td><td>48.5%</td></tr><tr><td>Very important</td><td>33.3%</td></tr></tbody></table> | Importance Level | Percentage | Not important | 0% | Slightly important | 18.2% | Important | 48.5% | Very important | 33.3% |
|----------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------|------------|---------------|----|--------------------|-------|-----------|-------|----------------|-------|
| Importance Level                 | Percentage                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |                  |            |               |    |                    |       |           |       |                |       |
| Not important                    | 0%                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |                  |            |               |    |                    |       |           |       |                |       |
| Slightly important               | 18.2%                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |                  |            |               |    |                    |       |           |       |                |       |
| Important                        | 48.5%                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |                  |            |               |    |                    |       |           |       |                |       |
| Very important                   | 33.3%                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |                  |            |               |    |                    |       |           |       |                |       |
| <b>Question</b>                  | <b>What improvements or additional features would you suggest for a sales forecasting system designed for small businesses?</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |                  |            |               |    |                    |       |           |       |                |       |
| <b>Aim of the Question</b>       | To collect user feedback and recommendations for system enhancement.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |                  |            |               |    |                    |       |           |       |                |       |
| <b>Findings &amp; Conclusion</b> | Responses were evenly distributed: easy way of approaching (25%), product-level categorization and forecasting (25%), mobile-friendly interface (25%), and positive feedback indicating satisfaction (25%). This demonstrates user interest in usability, mobility, and product-specific insights.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |                  |            |               |    |                    |       |           |       |                |       |

Table 11 - Survey Findings

### 4.5.3 Brainstorming/Observation

Used with key stakeholders (small business owners, IT staff and managers), brainstorming sessions and focus groups were the vital requirement elicitation method. These working sessions provided a melting pot of innovation combining technically feasible solutions with practical operating realities creating concepts to overcome constraints in traditional retail/inventory systems.

Here is an elaborated discussion of the key features identified:

1. **Dashboard Customization & Visualization:**

Stakeholders expressed the need for a user-friendly dashboard that displays graphical representations of sales forecasts and inventory trends. This feature allows non-technical users to easily interpret complex data and make informed business decisions. The ability to customize filters and export reports in different formats (PDF/Excel) for convenient reporting was also requested.

2. **Role-Based Access:**

Considering that multiple users (e.g., managers, analysts, and IT staff) will interact with the system, role-based access was identified as crucial to maintain data security and restrict users to only the necessary features and information.

## 4.6 Summary of Findings

As per the study, to facilitate inventory and financial management, SMEs require accurate and simple to comprehend sales forecasting systems. Even though the awareness about ML-based forecasting is moderate, the respondents appreciate the functionality of such systems, such as dashboards, visualization, and inventory tracking, and find them useful. The system will be welcomed and appreciated by the users to some degree of model transparency. To ensure security and usability, the stakeholders emphasized role-based access and customization of dashboards. At that, an easy-to-use machine-learning-based forecasting software against small business is clearly required.

## 4.7 Context Diagram

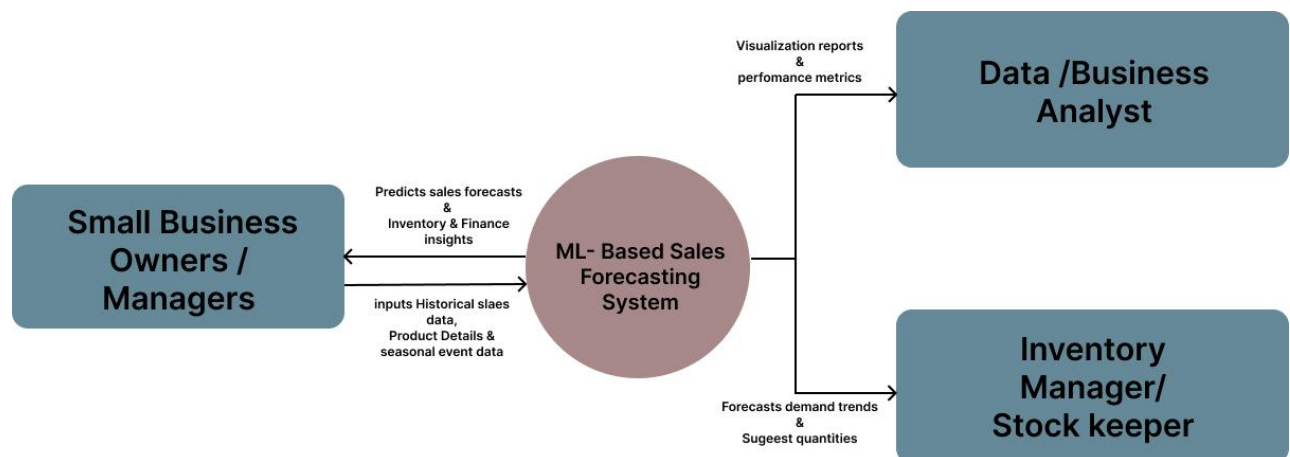


Figure 3 - Context Diagram

## 4.8 Use Case Diagram

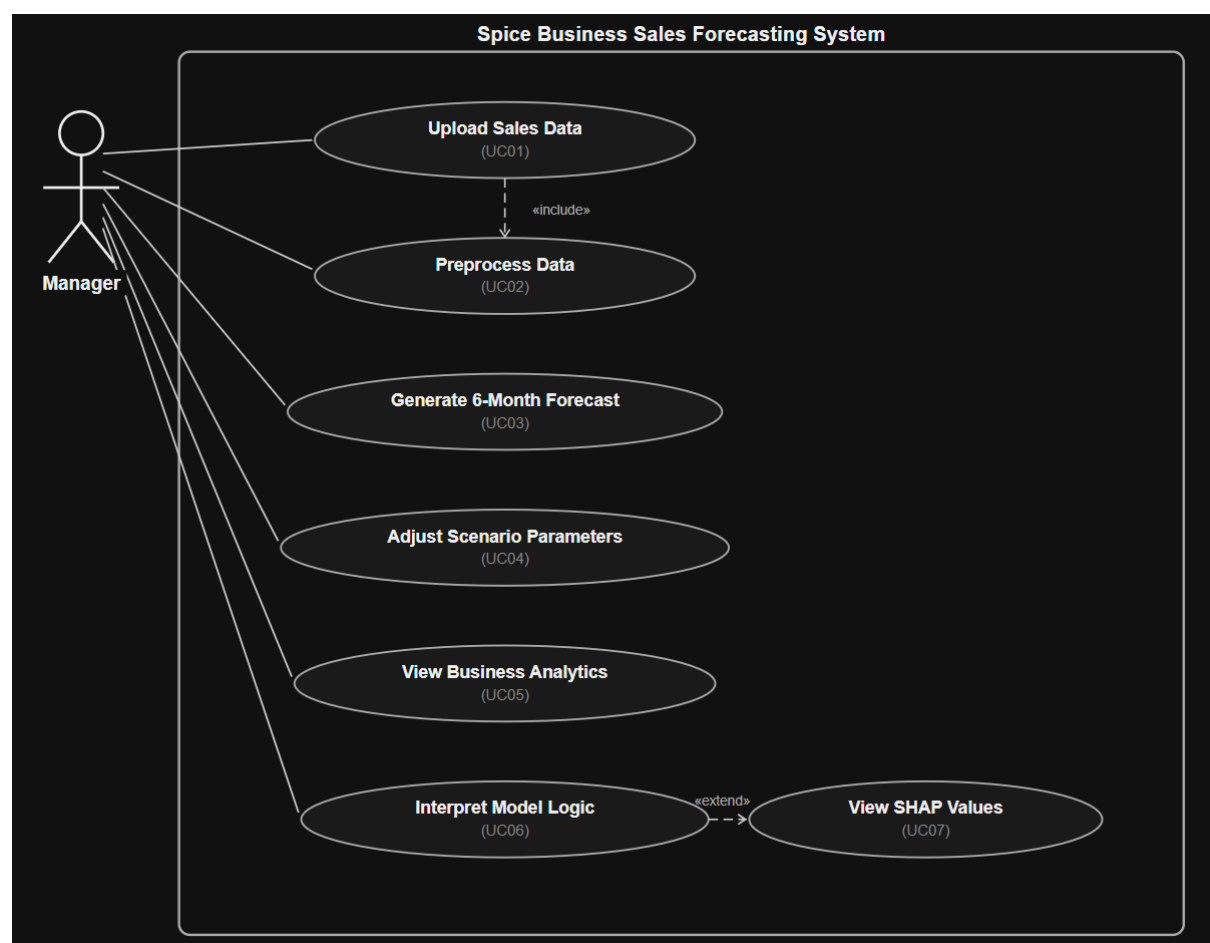


Figure 4 - Use Case Diagram

## 4.9 Use Case Descriptions

| Field          | Description                                                                                                                                                                                                                                         |
|----------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Use Case ID    | UC-01                                                                                                                                                                                                                                               |
| Use Case Name  | Upload Sales Data                                                                                                                                                                                                                                   |
| Actors         | Manager (Small Business Owner)                                                                                                                                                                                                                      |
| Goal           | To upload historical Excel/CSV sales files from the spice shop into the system for processing.                                                                                                                                                      |
| Preconditions  | The manager has access to the system. Raw sales data files are available in Excel or CSV format.                                                                                                                                                    |
| Postconditions | Sales data is successfully uploaded and stored, ready for preprocessing.                                                                                                                                                                            |
| Main Flow      | <ol style="list-style-type: none"> <li>1. Manager navigates to the data upload section.</li> <li>2. Manager selects and uploads the Excel/CSV file.</li> <li>3. System confirms the file has been received and queued for preprocessing.</li> </ol> |
| Field          | Description                                                                                                                                                                                                                                         |
| Use Case ID    | UC-02                                                                                                                                                                                                                                               |
| Use Case Name  | Preprocess Data                                                                                                                                                                                                                                     |
| Actors         | System (automated via data_processor.py)                                                                                                                                                                                                            |
| Goal           | To automatically clean, standardize dates, and handle missing values in the uploaded sales data.                                                                                                                                                    |
| Preconditions  | Raw sales data has been uploaded successfully (UC01 completed).                                                                                                                                                                                     |

| Postconditions | A cleaned, structured dataset is prepared and available for model training and forecasting.                                                                                                                                                                                                                                           |
|----------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Main Flow      | <ol style="list-style-type: none"> <li>1. System reads the uploaded raw file.</li> <li>2. System removes duplicates and handles missing values.</li> <li>3. System standardizes date formats and encodes categorical fields.</li> <li>4. Cleaned dataset is saved and flagged as ready for forecasting.</li> </ol>                    |
| Field          | Description                                                                                                                                                                                                                                                                                                                           |
| Use Case ID    | UC-03                                                                                                                                                                                                                                                                                                                                 |
| Use Case Name  | Manager (Small Business Owner)                                                                                                                                                                                                                                                                                                        |
| Actors         | Small Business Owner                                                                                                                                                                                                                                                                                                                  |
| Goal           | To produce a 6-month predictive sales forecast using the trained XGBoost regression model.                                                                                                                                                                                                                                            |
| Preconditions  | Cleaned historical sales data is available. The XGBoost model has been trained on that data.                                                                                                                                                                                                                                          |
| Postconditions | A visual and numerical 6-month sales forecast report is generated and displayed on the dashboard.                                                                                                                                                                                                                                     |
| Main Flow      | <ol style="list-style-type: none"> <li>1. Manager triggers the forecast from the dashboard.</li> <li>2. System feeds cleaned data and lag/rolling features into the XGBoost model.</li> <li>3. Model generates month-by-month revenue predictions.</li> <li>4. System displays the forecast as a chart and tabular report.</li> </ol> |
| Field          | Description                                                                                                                                                                                                                                                                                                                           |
| Use Case ID    | UC-04                                                                                                                                                                                                                                                                                                                                 |

|                |                                                                                                                                                                                                                                                                                                                                                            |
|----------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Use Case Name  | Adjust Scenario Parameters                                                                                                                                                                                                                                                                                                                                 |
| Actors         | Manager (Small Business Owner)                                                                                                                                                                                                                                                                                                                             |
| Goal           | To simulate different business scenarios by adjusting impact factors such as price changes or demand surges and observe their effect on the forecast.                                                                                                                                                                                                      |
| Preconditions  | A baseline 6-month forecast has already been generated (UC03 completed).                                                                                                                                                                                                                                                                                   |
| Postconditions | A revised forecast reflecting the adjusted scenario parameters is displayed for comparison.                                                                                                                                                                                                                                                                |
| Main Flow      | <ol style="list-style-type: none"> <li>1. Manager navigates to the scenario adjustment panel.</li> <li>2. Manager inputs impact factors (e.g., +10% price, seasonal demand surge).</li> <li>3. System recalculates the forecast using the modified parameters.</li> <li>4. Updated forecast is displayed alongside the baseline for comparison.</li> </ol> |
| <b>Field</b>   | <b>Description</b>                                                                                                                                                                                                                                                                                                                                         |
| Use Case ID    | UC-05                                                                                                                                                                                                                                                                                                                                                      |
| Use Case Name  | View Business Analytics                                                                                                                                                                                                                                                                                                                                    |
| Actors         | Manager (Small Business Owner)                                                                                                                                                                                                                                                                                                                             |
| Goal           | To view profit margins, top-performing products, and inventory requirements through an analytics dashboard.                                                                                                                                                                                                                                                |

| Preconditions  | Sales data has been uploaded and processed.<br>At least one forecast has been generated.                                                                                                                                                                                                                                    |
|----------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Postconditions | Key business metrics are displayed visually on the dashboard for decision-making                                                                                                                                                                                                                                            |
| Main Flow      | <ol style="list-style-type: none"> <li>1. Manager opens the analytics dashboard.</li> <li>2. System aggregates sales, profit, and inventory data.</li> <li>3. System displays charts for top products, profit trends, and stock requirements.</li> <li>4. Manager reviews insights to support business planning.</li> </ol> |
| Field          | Description                                                                                                                                                                                                                                                                                                                 |
| Use Case ID    | UC-06                                                                                                                                                                                                                                                                                                                       |
| Use Case Name  | Interpret Model Logic                                                                                                                                                                                                                                                                                                       |
| Actors         | Manager (Small Business Owner)                                                                                                                                                                                                                                                                                              |
| Goal           | To understand why the model made a specific prediction by viewing SHAP (SHapley Additive exPlanations) values for each feature.                                                                                                                                                                                             |
| Preconditions  | A forecast has been generated. SHAP analysis has been computed for the model output.                                                                                                                                                                                                                                        |
| Postconditions | A SHAP summary plot and feature importance rankings are displayed, explaining the model's reasoning.                                                                                                                                                                                                                        |

|           |                                                                                                                                                                                                                                                                                                                                                                             |
|-----------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Main Flow | <ol style="list-style-type: none"> <li>1. Manager selects a specific forecast period to inspect.</li> <li>2. System computes SHAP values for that prediction.</li> <li>3. System renders a SHAP bar chart showing each feature's contribution.</li> <li>4. Manager interprets which factors (e.g., lag features, rolling averages) most influenced the forecast.</li> </ol> |
|-----------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|

Table 12 - UseCase Description

## 4.10 Requirements

The MoSCoW principle is applied to determine the needs for this system according to its priority level. Both functional and non-functional requirements are categorised using the MoSCoW technique in the sections that follow.

### 4.10.1 Functional Requirements

| ID   | Requirement                                                 | Priority (MoSCoW) |
|------|-------------------------------------------------------------|-------------------|
| FR01 | Generate 6-month sales forecast (XGBoost)                   | Must Have         |
| FR02 | Upload and preprocess datasets (CSV/Excel).                 | Must Have         |
| FR03 | Fetching historical processed data via the /sales endpoint. | Must Have         |
| FR04 | Requesting a 6-month prediction using the XGBoost model.    | Must Have         |

|      |                                                              |             |
|------|--------------------------------------------------------------|-------------|
| FR05 | Generating the dashboard overview (Profit, Revenue, Trends). | Must Have   |
| FR06 | Identifying the highest-performing spice products.           | Should Have |

Table 13 - Functional Requirements

#### 4.10.2 Non-Functional Requirements

| ID    | Non-Functional Requirement | Description                                                                                                               | Priority (MoSCoW) |
|-------|----------------------------|---------------------------------------------------------------------------------------------------------------------------|-------------------|
| NFR01 | Performance                | The system must generate a sales forecast within an acceptable time frame (e.g., under 10 seconds for standard datasets). | Must Have         |
| NFR02 | Accuracy                   | The forecasting model should achieve acceptable predictive accuracy (e.g., RMSE or MAPE within industry standards).       | Must Have         |
| NFR03 | Usability                  | The interface should be intuitive and easily usable by non-technical business owners and managers.                        | Should Have       |

|       |                  |                                                                                                                                      |             |
|-------|------------------|--------------------------------------------------------------------------------------------------------------------------------------|-------------|
| NFR04 | Scalability      | The system should support the inclusion of new products, categories, or extended datasets without performance degradation.           | Should Have |
| NFR05 | Interpretability | The system should include SHAP-based visual explanations to make model predictions transparent and understandable for business users | Could Have  |

Table 14 - Non-functional Requirement

## 4.11 Chapter Summary

In this chapter, the SRS concerning this project is presented. This incorporated a stakeholder onion diagram to present the roles and association of the various stakeholders and a rich picture diagram to present the general framework of the system. In addition, This illuminated the various elicitation methods that were used to collect the system requirements and generalised the findings of every elicitation method. It also gave an elaborate description of how the system would be used and had recorded the functional and non functional requirements and prioritized them according to their significance.

## Chapter 5: Social, Legal, Ethical and Professional issues (SLEP)

### 5.1 Chapter overview

The chapter addresses the Social, Legal, Ethical, and Professional (SLEP) factors that are involved in the formulation and implementation of proposed Machine Learning-based Sales Forecasting System on a specialty retail spice shop. Given that the project works with real transactional sales information and produces predictive data, which can be used to make inventory planning, and cash flow decisions, care should be taken to make sure that the system follows responsible computing practices.

The development and implementation of this system were guided by the **BCS Code of Conduct (British Computer Society, 2021)**, which outlines professional responsibilities under four key principles:

1. The Public Interest
2. Professional Competence and Integrity
3. Duty to Relevant Authority
4. Duty to the Profession

These values were reconsidered and implemented across the project lifecycle to make sure that these factors followed social responsibility, legal requirements, and ethical behavior of AI, and professional ethics.

## **5.2 SLEP issues and mitigation**

The next part constitutes the discussion of the major Social, Legal, Ethical, and Professional concerns that were revealed in the course of the project and the ways they were alleviated.

### **5.2.1 Social Issues**

Automation of sales forecasting and inventory planning can help to decrease the role of a manual decision-making by the store staff. This can raise the issue of the lack of human intervention.

To counter this the system was made with a strict design of a decision supporting tool rather than to replace the employees. The ultimate decisions will be left with management. The outputs of the forecasts are shown in graphical and understandable forms to support the judgment of humans instead of dominating it. This is in accordance with the principle of the BCS of acting in the Public Interest, which makes technology beneficial to the stakeholders.

A second social issue is associated with data bias and poor coverage. As only 1 year of data has been used to train the model on the data of a single retail shop, generalizability can be limited.

This was a limitation that was recorded. It was scoped to be used in a single-store manner and no one was making any statement of the extended applicability in a retail manner. This provides accountability and openness in communicating capabilities of the systems.

### **5.2.2 Legal Issues**

The project is based on real sales data of a retail business in the form of transactions. Despite the fact that no customer-identifiable information was employed, the records of sales are sensitive business data.

To address data protection and confidentiality concerns, several measures were implemented. No personally identifiable information (PII) was collected or used within the system. The dataset was anonymised prior to model development to ensure privacy. Permission was

obtained from the business owner before using the data, and the dataset was handled securely without being shared or made publicly accessible.

These measures allow adhering to the principles of data protection and proper data management. The stakeholder feedback was taken during the requirement elicitation. Any information that can be identified must be informed according to the standard of ethics in research. To address ethical concerns related to participant involvement, appropriate measures were implemented. Interview responses were anonymised to protect participant identity, and no personal information was disclosed without prior consent. Participants were clearly informed about the purpose of the research before their involvement, ensuring transparency and ethical compliance.

### **5.2.3 Ethical Issues**

The XGBoost machine learning models are also perceived to be black-box systems. When predictions make a difference in financial decisions and this is not transparent, ethical issues can come about.

In response to this, SHAP (SHapley Additive Explanations) was added to improve the interpretability. Importance of features was also given to show how the predictions were made. This justifies the use of AI ethically and fits in the BCS codes of conduct related to integrity and transparency.

The use of automated forecasts can cause over-reliance on the part of the user and this can end up in inaccurate decision making in case decision users fail to critically analyze the prediction. To avoid such a situation, common measures of evaluation like MAE, RMSE, and MAPE are presented to show the robustness of the predictions. Moreover, the results of prediction are shown in a transparent and understandable format. It also invites users to use model predictions along with domain knowledge and hands-on experience in making business decisions. Also, integrity in the research was ensured through honest reporting of research findings, not manipulating data, and using relevant cross-validation.

### 5.2.4 Professional Issues

The creation of the ML-based forecasting system presupposes the skills in the data preprocessing, time series modelling, and statistical evaluation. Creating untrustworthy products because of a lack of knowledge would be a violation of professionalism.

In order to partially eliminate possible risks and guarantee the validity of the study, the extensive literature review was performed to find appropriate practices and methods. Several machine learning models were tested or compared in a similar fashion with the help of consistent benchmarking procedures to give a fair and accurate results. Model performance was verified with standard assessment metrics and the whole experience of developing and assessing the model was well documented to ensure transparency and replicability.

Academic integrity was also maintained by citing it properly, implementing it originally and reporting it clearly on limitations of the system.

### 5.2.5 SLEP Summary Table

| Social                                                                                                                                                                                                    | Legal                                                                                                                                                                                                                                    |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <ul style="list-style-type: none"> <li>● Reduced reliance on manual decision-making.</li> <li>● Risk of dataset bias (single-store data).</li> <li>● Responsible AI deployment in SME context.</li> </ul> | <ul style="list-style-type: none"> <li>● Protection of business sales data.</li> <li>● Anonymization of records.</li> <li>● Informed consent for stakeholder feedback.</li> <li>● Compliance with data protection principles.</li> </ul> |
| Ethical                                                                                                                                                                                                   | Professional                                                                                                                                                                                                                             |

|                                                                                                                                                                                                  |                                                                                                                                                                                                                                         |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <ul style="list-style-type: none"> <li>● Black-box nature of ML models.</li> <li>● Risk of blind trust in automated forecasts.</li> <li>● Need for transparent reporting of accuracy.</li> </ul> | <ul style="list-style-type: none"> <li>● Adherence to BCS Code of Conduct.</li> <li>● Maintenance of academic integrity.</li> <li>● Competent implementation and benchmarking.</li> <li>● Proper documentation and citation.</li> </ul> |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|

*Table 15: SLEP Summary Table*

### 5.3 Chapter summary

The chapter has explored the Social, Legal, Ethical, and Professional (SLEP) issues related to the development of the Sales Forecasting System based on machine learning in a specialty retail spice shop. Considering the fact that the system deals with real business data and that the system will affect the financial and inventory decisions, the appropriate system design and deployment were mandatory.

In the social context, the system was placed as a decision support tool so as to see that it complements the human decision making and not to substitute it. The legal issues of business data confidentiality and consent of stakeholders were discussed via anonymization, limited data sharing practices and informed consent. The lack of transparency in models, and excessive use of automated predictions, were addressed by including explainability methods (SHAP) and properly reporting on the performance metrics. In terms of profession, the project complied with the BCS Code of Conduct since it was technically competent and academically honest and documentation was transparent.

Comprehensively, in the present project, the fact that sophisticated predictive analytics could be applied in a rather responsible way in a small retail setting, when the right precautions, transparency and professional frameworks are observed

## Chapter 6 : Design

### 6.1 Chapter Overview

In this chapter, the system design of the proposed Machine Learning based Sales Forecasting and Inventory Planning System is outlined that was created to address the needs of a specialty retail spice shop. The design phase will convert the requirements established in Chapter 4 (Software Requirements Specification) into a formal structure of architectural and component-level map that will be used to implement it.

The design stage is aimed at making sure that the system architecture, the data flow and the modelling elements are designed in a way that fulfills both the non-functional requirements (e.g. performance, accuracy, usability, maintainability) and functional requirements (e.g. sales forecasting, inventory estimation, performance evaluation).

The design process entails the identification of the overall system architecture besides making sure that everything works very well together. It involves the development of a powerful data processing and machine learning pipeline capable of managing data preparation, training of the model and prediction. Moreover, it entails the design of model evaluation and explainability elements to evaluate performance and give an insight into model behavior. The process also provides alignment to the system-wide quality attributes like scalability, reliability and efficiency as well as converting the research objectives to the practical and implementable modules in the system.

The chapter thus shows how the proposed design is capable of supporting the research purpose to develop a valuable and interpretable system along with a 6 month forecasting system that is business oriented and specific to SME retail settings.

### 6.2 Design Goals

The design objectives determine the important quality characteristics that guide the system architecture and implementation. They are direct objectives based on the Non-Functional Requirements (NFRs) of the SRS chapter and applied to all the parts of the system.

#### 1. Accuracy

Goal:

- The system should be able to produce the sales forecasts that are reliable and statistically proven to assist in making inventory and financial planning decisions.

Design Strategy:

- Application of managed machine learning models (XGBoost and comparative models).
- Application of feature engineering such as time-based variables and cyclic encoding.
- Performance appraisal of MAE, RMSE, MAPE and R<sup>2</sup>.
- Use of cross-validation method to guarantee temporal integrity.
- Honesty is the key of the research purpose and the quality of inventory and cashflow analysis.

## 2. Performance Efficiency

Goal:

- The system should have the ability to provide predictions in reasonable time limits without incurring too much computation.

Design Strategy:

- XGBoost is used because it is computationally efficient with structured tabular data.
- Selective dimensionality to remove unnecessary dimensions.
- Serialization of model (pickle format) to prevent retraining on prediction stage.
- 6-month projections made using batch forecasting method.
- This makes the system viable to the SME setting having fewer computing capabilities.

## 3. Usability

Goal:

- Forecasts should be easily interpreted and made understandable to non technical stakeholders in retailing.

Design Strategy:

- Graphical plotting- charting of historical sale and forecasted sales.
- Forecast results in tabular form and summary form.
- SHAP-based explainability was integrated to understand feature importance.
- Good presentation of evaluation metrics.
- The design will make the system a decision-support tool and not a strictly technical model of ML.

## 4. Interpretability

Goal:

- The model has to offer clear explanations to its predictions.

Design Strategy:

- SHAP (SHapley Additive Explanations) integration.
- Visualization using the ranking of feature importance.
- Determination of seasonal and trend effect on prediction.
- It is a direct contribution to the principles of ethical AI and it is aligned with the objective of the research explainable forecasting.

## 6.3 System Architecture Diagram

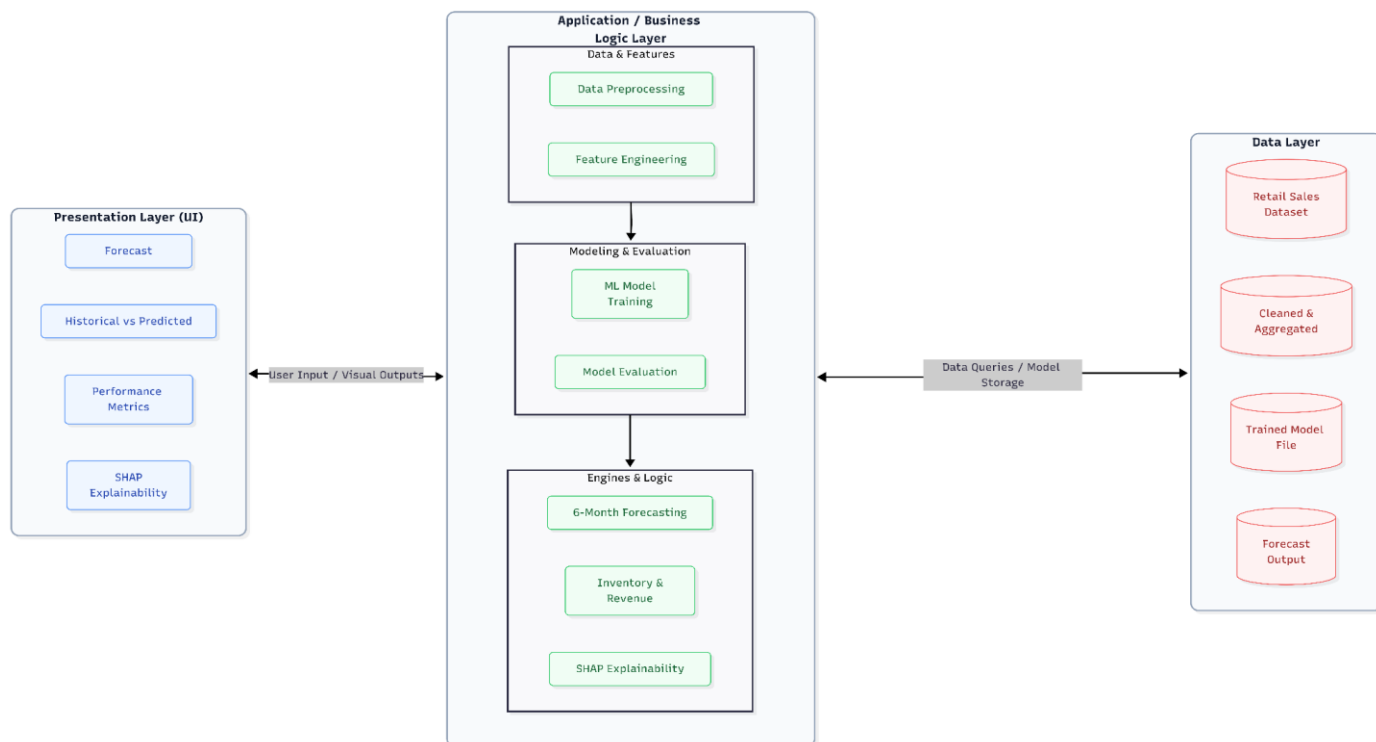


Figure 7: System Architecture Diagram

The architecture consists of three primary layers:

1. Presentation Layer
2. Application / Business Logic Layer
3. Data Layer

This structure supports maintainability, scalability, and clear integration of forecasting, evaluation, and explainability components.

#### 1. Presentation Layer (UI)

The Presentation Layer is in charge of the interaction with the user and visualization of the results.

The design process entails the identification of the overall system architecture besides making sure that everything works very well together. It involves the development of a powerful data processing and machine learning pipeline capable of managing data preparation, training of the model and prediction. Moreover, it entails the design of model evaluation and explainability elements to evaluate performance and give an insight into model behavior. The process also provides alignment to the system-wide quality attributes like scalability, reliability and efficiency as well as converting the research objectives to the practical and implementable modules in the system.

This layer provides usability and interpretability so that the retail stakeholders can interpret predictions without technical expertise. The system is a decision support system, and not an automated decision-maker.

#### 2. Application / Business Logic Layer.

This is the heart of the processing of the system and it is subdivided into three major segments:

##### a) Data & Feature Processing

- Data preprocessing
- Cleaning and aggregation
- Feature engineering

##### b) Modelling & Evaluation

- Training of ML models (XGBoost and comparative models)
- Statistical evaluation of the model.
- Cross-validation

#### c) Engines & Business Logic

- 6-Month Forecasting Engine
- Inventory Revenue Estimation Logic.
- SHAP Explainability Engine

This layer has given the research purposes a reality of converting past sales records into readable future predictions and business information.

### 3. Data Layer

Data Layer Data Layer contains and maintains all the data components in the system:

- Retail sales dataset
- Cleaned and aggregated data
- Trained model file
- Forecast output results

It provides integrity of data, organization of data and restricted access. Model serialization enables trained models to be reused during prediction.

## 6.4 Detailed Design

This part will provide the detailed design of SalesSense system in terms of Object-Oriented Analysis and Design (OOAD). The detailed design is intended to establish the internal structure, interaction between components and how the system will operate above the high-level architecture that is provided in Section 6.3.

The following design diagrams are included:

- Activity Diagram
- Sequence Diagram
- Class Diagram
- Data Processing & Model Pipeline Diagram

Each diagram represents a different perspective of the system.

### 6.4.1 Activity Diagram

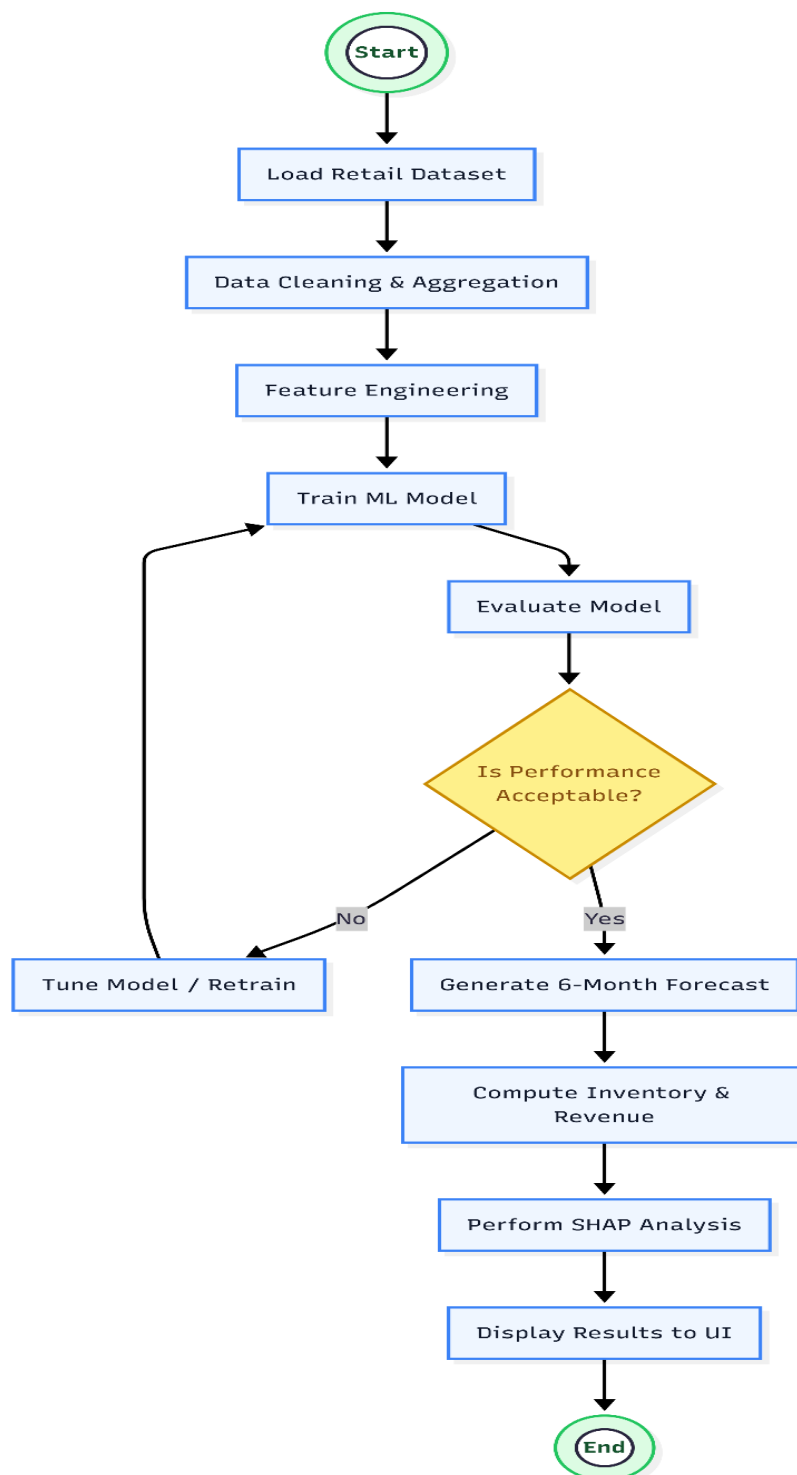


Figure 8:Activity Diagram

The Activity Diagram shows all the workflow of the forecasting system which starts with data loading and ends with the visualization of the final result. It is a logical control flow and decision-making process in the system.

It can start by loading the retail data, then cleaning and aggregating the data and engineering features. Training and evaluation of the ML model is then done. In case the performance is unsatisfactory, hyperparameter optimization and re-training is done. After the satisfactory performance has been obtained, the system creates a 6-month forecast, calculates inventory and revenue projections, conducts SHAP analysis, and presents the final results.

This has been illustrated in this diagram showing how model optimization is an iterative process and one that has to be accurate before deployment.

6.4.2 Sequence Diagram

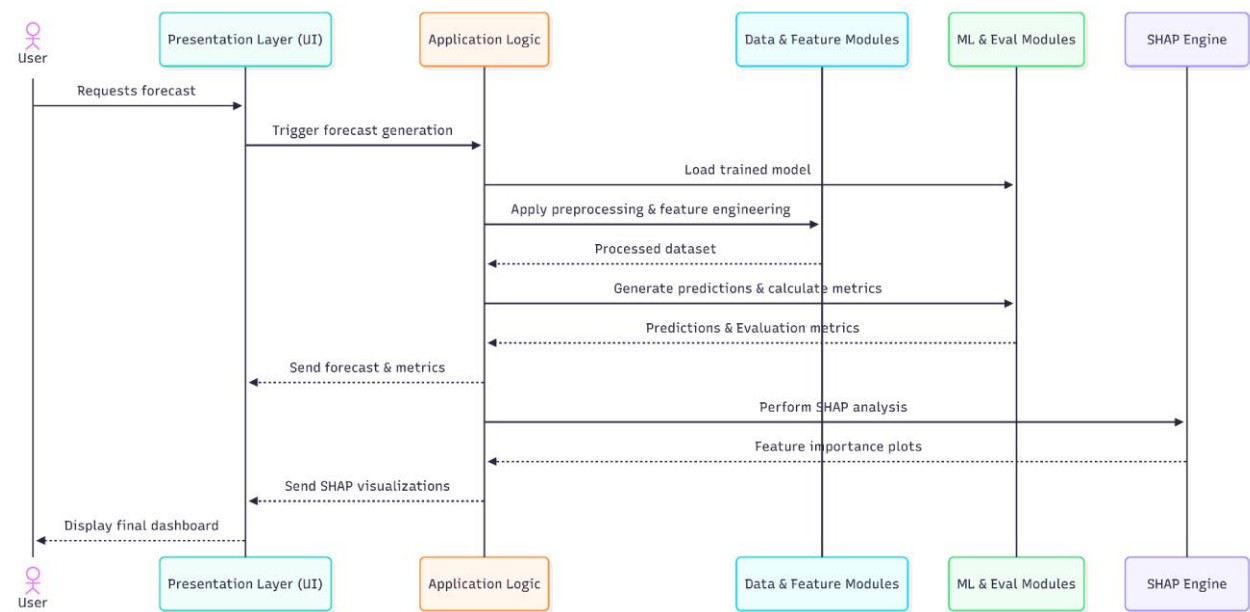


Figure 9:Sequence Diagram

The Sequence Diagram shows the interactions between components of the system with time as a user requests a forecast.

The flow of interactions involves:

1. Forecasting via User Interface.
2. Forecast generation is initiated by application logic.
3. Trained model is loaded
4. Preprocessing and feature engineering are used.
5. Predictions and measures of evaluation are produced.
6. SHAP analysis is performed
7. The outcomes of the forecast and visualizations are sent back to the UI.

This diagram identifies only the runtime communication with the modules and ensures a tight separation of presentation, business logic, data modules and explainability engine.

### 6.4.3 Class Diagram

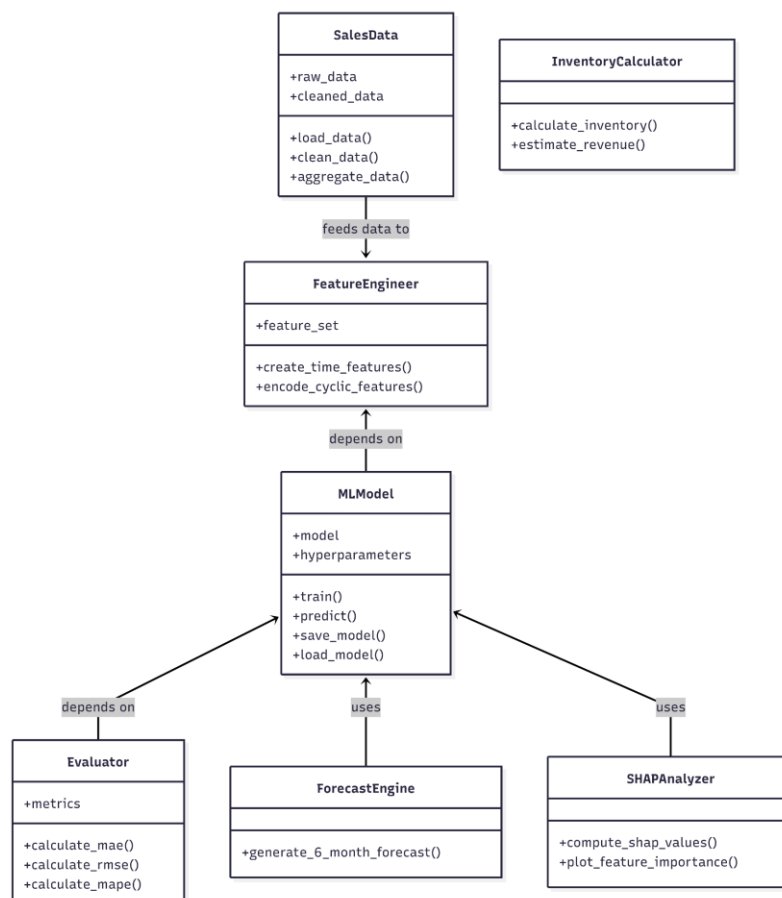


Figure 10:Class Diagram

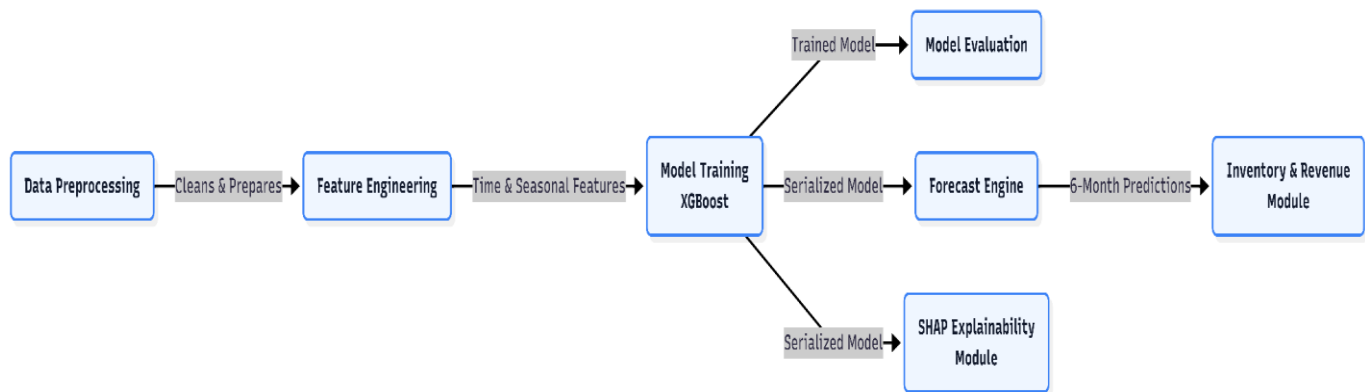
The Class Diagram establishes the object-oriented design of the system through the determination of core classes, attributes and methods.

Key classes include:

1. SalesData - Processes the data loading, cleaning and aggregation.
2. FeatureEngineer - Generates time and seasonal features.
3. MLModel -Training, prediction, saving, and loading.
4. Evaluator -Computes MAE, RMSE and MAPE.
5. ForecastEngine - 6-month forecasting.
6. InvCalc- Predicts inventory and turnover.
7. SHAPAnalyzer - Does feature importance analysis.

Modularity and maintainability is guaranteed by the relationships between classes. The MLModel is the core elements of forecasting, evaluation, and explainability modules.

#### 6.4.4 Data Processing & Model Pipeline Diagram



*Figure 11 : Data Processing & Model Pipeline Diagram*

This graph shows the internal machine learning pipeline and data transformation process.

The pipeline consists of:

Data Preprocessing → Feature Engineering → Model Training (XGBoost) → Model Evaluation → Forecast Engine → Inventory & Revenue Module → SHAP Explainability

The trained model is saved and reused to generate predictions and explanations, forming an organized pipeline that ensures consistent and reliable operation. This approach guarantees reproducibility by allowing the same results to be obtained across different runs, improves efficiency by avoiding repeated training, and enables clear decoupling between the training and prediction phases. Additionally, it supports informed business decision-making by providing reliable forecasts and interpretable insights.

## 6.5 Algorithm Design

The core of the forecasting system relies on the XGBoost (Extreme Gradient Boosting) algorithm for revenue prediction and a linear trend analysis for inventory forecasting. This section details the algorithmic approach used to transform raw data into actionable insights.

```
ALGORITHM: SalesAndInventoryForecasting
INPUT: Raw Sales Data (CSV, Excel files containing 'Particulars', 'Quantity', 'Value')
OUTPUT: 6-Month Revenue Forecast, Next Month Inventory Recommendations
// PART 1: Data Ingestion & Cleaning (ETL)
FUNCTION ProcessData(directory):
    Initialize empty collection `CleanedData`

    FOR EACH file in directory DO:
        Locate Header Row (Find "Particulars")
        Map Date Columns based on headers

        FOR EACH Row in file DO:
            Extract `Product Name`
            IF row is valid Product THEN:
                FOR EACH `Date` in map DO:
                    Extract `Quantity` and `Value`
                    Clean values (Remove "kgs", ",", handles NaNs)
                    Add {Date, Product, Quantity, Value} to `CleanedData`
                END FOR
            END IF
        END FOR
    END FOR

    RETURN `CleanedData`
// PART 2: Forecasting Engine
FUNCTION GenerateForecast(CleanedData):
    // A. Financial Forecasting (XGBoost)
    AggregatedData = Group `CleanedData` by Month (Sum of Value)
```

Add Feature `TimeIndex` (0, 1, 2 ... N) representing months

Split Data into Features (X) and Target (y)

Initialize XGBRegressor:

- Objective: 'reg:squarederror'
- N\_Estimators: 100
- Learning\_Rate: 0.05

Train Model(X, y)

Generate `FutureIndices` for next 6 months

`RevenueForecast` = Model.Predict(`FutureIndices`)

// B. Inventory Recommendation (Product-Level)

`TopProducts` = Identify Top 5 Products by Total Revenue

Initialize `InventoryPlan`

FOR EACH `Product` in `TopProducts` DO:

    Get Monthly Quantity History

    // Calculate Trend Slope

    Slope = (LastQuantity - FirstQuantity) / TotalMonths

    `PredictedQty` = LastQuantity + Slope

    Determine Status:

        IF `PredictedQty` > CurrentAverage THEN

            Status = "Stock Up"

        ELSE

            Status = "Sufficient"

    Add to `InventoryPlan`

END FOR

RETURN `RevenueForecast`, `InventoryPlan`

6.6 UI Design

This section outlines the layout and design decisions for the frontend dashboard. The interface is designed to be intuitive, ensuring that complex machine learning insights are accessible to non-technical users (business owners).

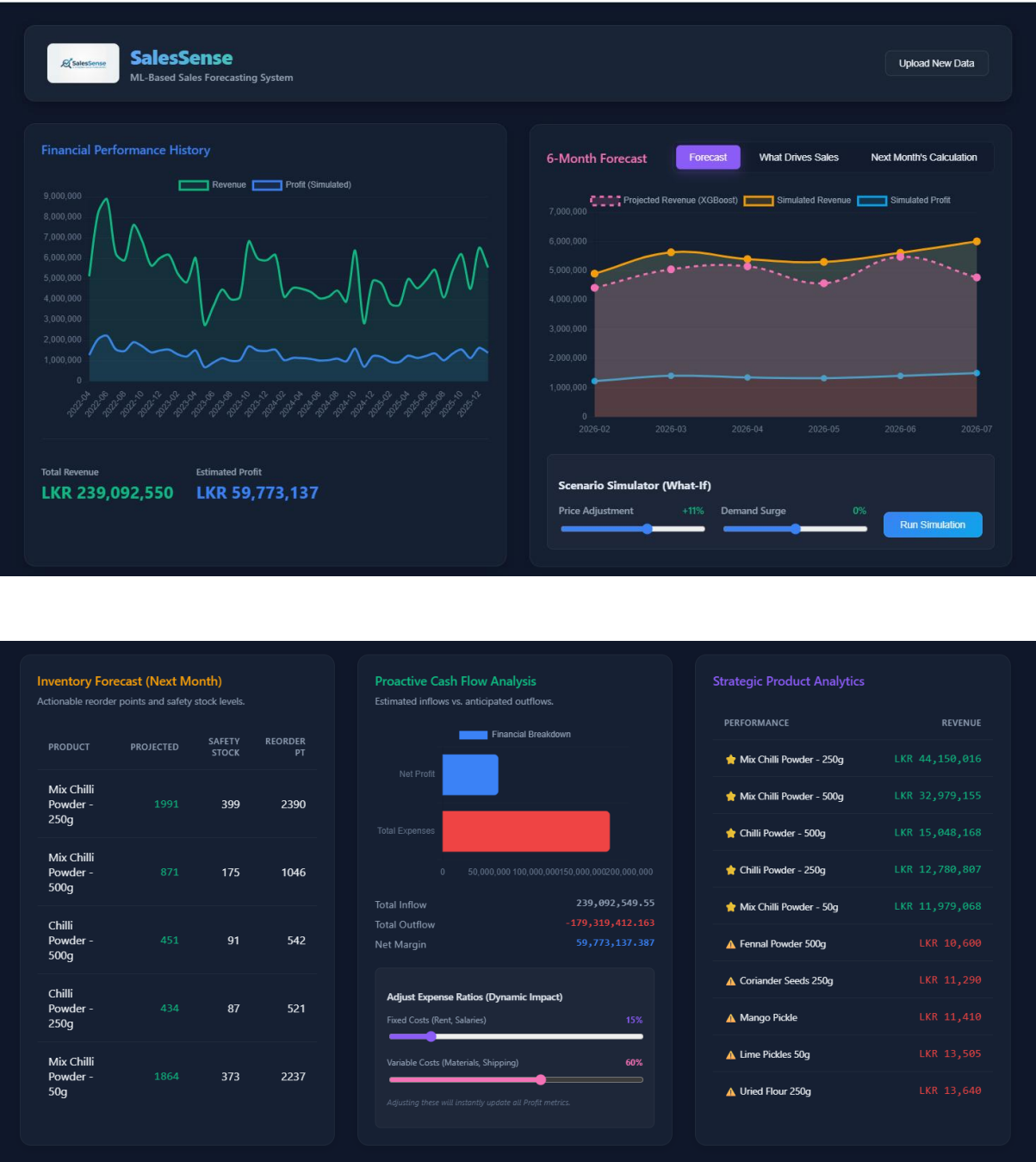


Figure 11 : UI Diagram

## 6.7 Chapter Summary

This chapter contained the entire system design of SalesSense forecasting system. The design stage converted the non-functional and functional requirements into a layered architecture structure that comprised presentation, business logic, and data layers. Object-Oriented Analysis and Design (OOAD), such as the activity diagram, sequence diagram, class diagram, and machine learning pipeline diagram, were introduced to design the detailed diagrams. These drawings showed system workflow, component interaction, object structure and data transformation processes. The algorithm design part formally outlined the hybrid forecasting model which was the combination of XGBoost to predict revenue and linear trend to predict inventory. This design is accurate, can be interpreted, has modularity and scaleable and yet can be deployed by SME. In general, the system design offers a solid implementation basis and helps to meet the research goals of the accurate, explainable, and business-oriented sales forecasting.

## Chapter 7: Implementation

### 7.1 Chapter Overview

This chapter describes the implementation of the proposed SalesSense forecasting system, translating the design and algorithmic structures presented in Chapter 6 into a functional prototype. The implementation phase focused on developing modular components for data preprocessing, feature engineering, XGBoost-based revenue forecasting, inventory recommendation logic, and SHAP-based explainability. Following the layered architecture defined earlier, each module was developed and integrated to ensure accuracy, maintainability, and usability. This chapter outlines the technologies used and explains how the system design was transformed into a working machine learning application.

### 7.2 Technology Selection

This section justifies the selection of tools, frameworks, and programming languages used for the development of the prototype and the AI/ML model. The choices were made to ensure high performance, scalability, and ease of integration between the machine learning backend and the interactive frontend.

#### 7.2.1 Technology Stack

The following figure visualizes the technology stack, separated into the Frontend (User Interface), Backend (API & Processing), and Data/ML Layer.

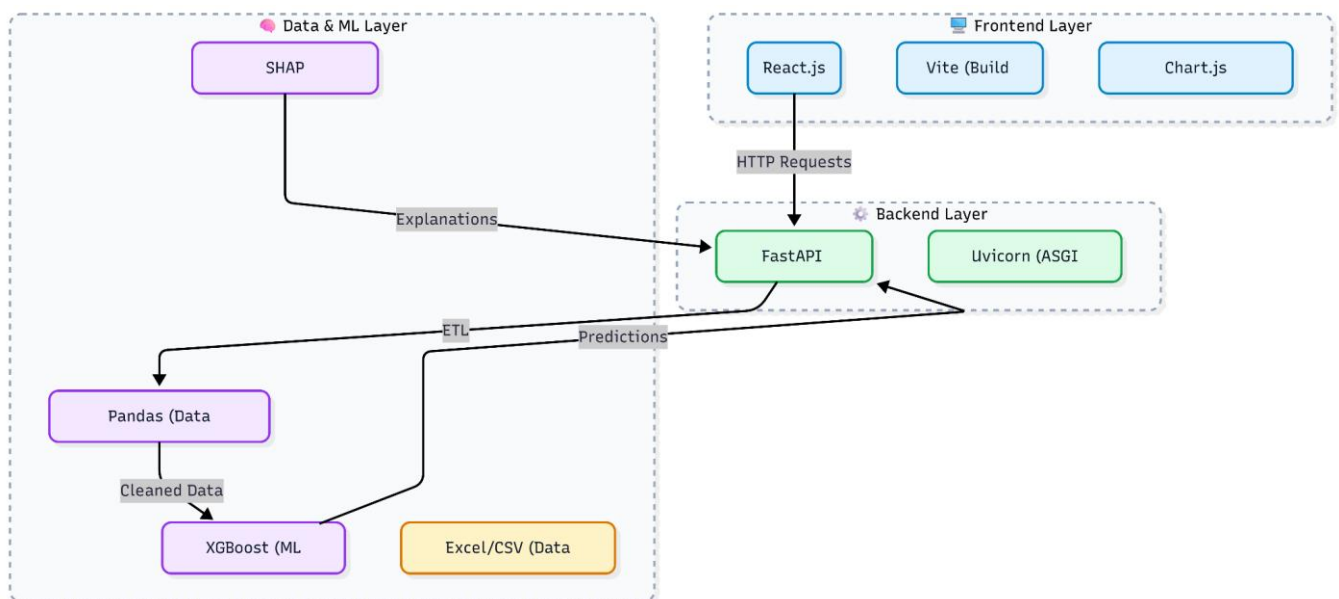


Figure 12 : Technology Stack

### 7.2.2 Programming Languages

**Python:** Selected as the primary language for the backend and machine learning components.

Python is the industry standard for Data Science and AI due to its extensive ecosystem of libraries (Pandas, Scikit-learn). Its simplicity allows for rapid prototyping of complex algorithmic logic.

**JavaScript (ES6+):** Used for the frontend development.

JavaScript is the native language of the web, enabling dynamic, client-side interactivity essential for a responsive dashboard.

**CSS3:** Used for styling the user interface.

Modern CSS variables and Flexbox/Grid layouts were used to create a custom "Glassmorphism" aesthetic without the overhead of heavy UI frameworks.

### 7.2.3 Development Framework

**FastAPI (Backend):** A modern, high-performance web framework for building APIs with Python.

Chosen over Flask or Django for its speed (Starlette-based) and automatic validation using Python type hints. It natively supports asynchronous processing, which is beneficial for handling ML inference requests concurrently.

**React.js (Frontend):** A component-based JavaScript library for building user interfaces.

React's Virtual DOM ensures efficient rendering of dynamic charts and data tables. Its component-based architecture makes the code modular and reusable (e.g., reusing the same Chart component for different metrics).

### 7.2.4 Libraries/ Toolkits

1. **XGBoost:** The core machine learning library for regression forecasting.
  - Justification: XGBoost (Extreme Gradient Boosting) was selected for its superior performance on structured/tabular data compared to Deep Neural Networks, which often require more data than is available in this context. It handles missing values and non-linear relationships effectively.
2. **SHAP (SHapley Additive exPlanations):** used for model interpretability.
  - Justification: Crucial for trust. It allows the system to explain why a forecast was made, moving beyond "black box" predictions.

3. Pandas & NumPy: Used for data manipulation and numerical analysis.
  - Justification: Essential for the ETL (Extract, Transform, Load) process to clean raw Excel files and structure them for the model.

### 7.2.5 Integrated Development Environment (IDEs)

**Visual Studio Code (VS Code):** The primary IDE used for full-stack development.

Justification: VS Code offers excellent support for both Python and JavaScript ecosystems via extensions. Features like the integrated terminal, IntelliSense (code completion), and debugging tools streamlined the development workflow across both frontend and backend.

### 7.2.6 Summary of the Technology Selection

The technology selection prioritizes **performance** and **maintainability**. Python and XGBoost provide a robust engine for accurate forecasting, while FastAPI ensures these insights are served with low latency. On the client side, React and Chart.js deliver a seamless, app-like experience. This decoupled architecture (separate Frontend and Backend) allows for independent scaling and maintenance of the user interface and the predictive core.

## 7.3 Core Functionalities Implementation

This section details how the core features of the **SalesSense** system were developed, covering data analysis, modular implementation, and system integration.

### 7.3.1 Exploratory Data Analysis (EDA)

- Before training the model, an exploratory analysis was conducted on the dataset (generated from files/\*.xlsx and april.csv) to understand sales distribution and identify data quality issues.
- Dataset Declaration: The system utilizes a proprietary dataset of monthly sales records from a retail spices sales shop.
- Dataset Size and Time Range: The dataset consists of 43 usable monthly records covering the period from April 2022 to January 2026. The first three months were excluded during preprocessing due to insufficient lag values required for feature engineering.
- Data Split: Training Data (40 Months): Used to teach the XGBoost model the relationship between time (Month) and Revenue.
- Testing Data (3 Months): Used to validate the model's accuracy on unseen future months.
- Features Used: The final model was trained using the following features: MonthNum, MonthOfYear, Lag1, Lag2, Lag3, RollingMean3, and RollingStd3. These features were specifically engineered to capture temporal dependencies and seasonal patterns in sales data.

➤ Feature Engineering:

To improve predictive performance, several time-series-based features were generated from the original dataset. Lag features (Lag1, Lag2, Lag3) were created to capture short-term dependencies between consecutive months. Rolling statistics such as RollingMean3 and RollingStd3 were introduced to represent local trends and variability in sales. Additionally, MonthNum and MonthOfYear were included to capture seasonality effects. These engineered features enable the model to better learn non-linear patterns in the data.

### 7.3.2 Modular Development

The system is built using a modular architecture, ensuring that data processing, training, and inference are loosely coupled.

#### A. Data Preprocessing Module (data\_processor.py)

This module handles the "Extract, Transform, Load" (ETL) process. It reads raw, unstructured Excel files and converts them into a clean CSV format.

Key Implementation: The process\_dataframe function scans for the "Particulars" row to identify the start of the data table, handling dynamic header positions found in manual bookkeeping files.

```
def clean_value(val):  
    """  
    Cleans currency and unit strings (e.g., '1,200', '10 kgs')  
    into float values for analysis.  
    """  
    if pd.isna(val):  
        return 0.0  
    s = str(val).replace(",", "").strip()  
    match = re.search(r"([\d\.]+)", s)  
    if match:  
        return float(match.group(1))  
    return 0.0
```

## B. Prediction Engine (main.py)

This module wraps the XGBoost model within a FastAPI endpoint. It trains the model on-the-fly when a request is received, ensuring the forecast always reflects the latest data.

- **Model Training Process:** The dataset was split into training and testing sets, where approximately 80% of the data (40 months) was used for training and the remaining 20% (3 months) was used for testing. The XGBoost regressor was trained on the training dataset using the engineered features, and predictions were generated for the test period as well as future months.
- **Key Implementation:** The `/api/forecast` endpoint prepares the data, trains the XGBRegressor, and generates future indices for prediction.
- **Model Optimization:** Basic hyperparameter tuning was applied to improve model performance and reduce overfitting. Parameters such as learning rate, number of estimators, and maximum tree depth were adjusted. A lower learning rate was used to ensure gradual learning, while limiting tree depth helped prevent the model from capturing noise in the small dataset. These adjustments improved generalization on unseen data.
- **Model Evaluation Metrics:** The performance of the model was evaluated using standard regression metrics, including Root Mean Squared Error (RMSE), Mean Absolute Error (MAE), Mean Absolute Percentage Error (MAPE), and R-squared ( $R^2$ ). These metrics provide insights into prediction accuracy and model reliability. Detailed evaluation results are presented in Chapter 8.
- **Model Explainability using SHAP:** To enhance transparency, SHAP (SHapley Additive Explanations) was integrated into the system to interpret model predictions. SHAP values quantify the contribution of each feature toward the final prediction, allowing users to understand which factors influence sales forecasts. This improves trust in the model and supports better decision-making for non-technical users.

```
@app.get("/api/forecast")
def get_forecast(model_type: str = "xgboost"):
    # ... data loading ...

    model = xgb.XGBRegressor(
        objective='reg:squarederror',
        n_estimators=100,
        learning_rate=0.05
    )
    model.fit(X, y)
    future_preds = model.predict(future_X)

    # ... returns JSON response ...
```

```

# XGBoost Model
model = xgb.XGBRegressor(
    objective='reg:squarederror',
    n_estimators=100,
    random_state=42
)
model.fit(X, y)

# Iterative 6-Month Forecast
future_preds = []
last_known = monthly_data.copy()
first_month_X = None

for i in range(1, 7):
    next_date = last_date + pd.DateOffset(months=i)
    roll_vals3 = last_known["Value"].iloc[-3:]

    next_X = pd.DataFrame([
        {"MonthNum": last_month_num + i,
         "MonthOfYear": next_date.month,
         "Lag1": roll_vals3.iloc[-1],
         "Lag2": roll_vals3.iloc[-2],
         "Lag3": roll_vals3.iloc[-3],
         "RollingMean3": roll_vals3.mean(),
         "RollingStd3": roll_vals3.std()}
    ])

    if i == 1:

```

### C. Inventory Logic Module

Integrated within the forecasting endpoint, this logic calculates the "Rate of Sale" for top products to predict next month's demand.

```

# Logic for Inventory Recommendation
if len(p_monthly) > 1:
    # Calculate slope to determine trend direction
    slope = (p_monthly["Quantity"].iloc[-1] - p_monthly["Quantity"].iloc[0]) /
len(p_monthly)
    next_qty = p_monthly["Quantity"].iloc[-1] + slope

```

### 7.3.3 Code Structure & Organization

The codebase follows a clear separation of concerns pattern:

- 1.backend/: Contains all business logic and ML algorithms.
- 2.main.py: The Controller, handling API requests.
- 3.data\_processor.py: The Service, handling data logic.
- 4.frontend/: Contains the presentation layer.
- 5.src/App.jsx: The View, responsible for rendering charts and managing state.
- 6.Design Pattern: The backend implements a Service-Controller pattern (implicitly), where
- 7.Data\_processor.py acts as the service layer providing clean data to the main.py controller. The frontend uses the Component-Based pattern inherent to React.

### 7.3.4 Integration

Integration between the Frontend and Backend is achieved via RESTful APIs.

- 1.Request: The React frontend uses axios to send an HTTP GET request to `http://localhost:8000/api/forecast`.
- 2.Processing: FastAPI receives the request, triggers the training pipeline in `main.py`, and generates the JSON response.
- 3.Response: The JSON data (containing predictions and SHAP values) is sent back to the frontend.
- 4.Rendering: React updates the state (`setForecast`), causing the `react-chartjs-2` components to re-render with the new data.

This seamless integration allows the complex ML operations to occur invisibly to the user, who simply sees the dashboard update in real-time.

## 7.4 User Interface Implementation

This section describes the implementation of the user interface (UI) of the SalesSense system and its integration with backend forecasting modules. The UI was developed as a simple dashboard-based interface to enable retail stakeholders to generate forecasts, view model performance, and interpret results without requiring technical knowledge.

The user interface was implemented as a structured dashboard that integrates revenue forecasting, model evaluation, inventory recommendations, and explainability into a unified decision-support platform. The frontend includes a forecast trigger button, a six-month revenue summary panel, historical vs predicted line chart visualizations, model performance metrics (MAE, RMSE, MAPE), and a SHAP-based feature importance section to enhance transparency. The frontend communicates with backend modules through structured function calls, where the trained XGBoost model is loaded, preprocessing and feature engineering are applied, forecasts are generated, inventory recommendations are computed, and explainability outputs are produced before returning results to the UI. Model serialization ensures efficient runtime execution without retraining. Usability and accessibility were prioritized through clear layout design, readable fonts, high contrast, minimal technical terminology, and avoidance of color-only visual cues, ensuring the system remains understandable and accessible for SME stakeholders.

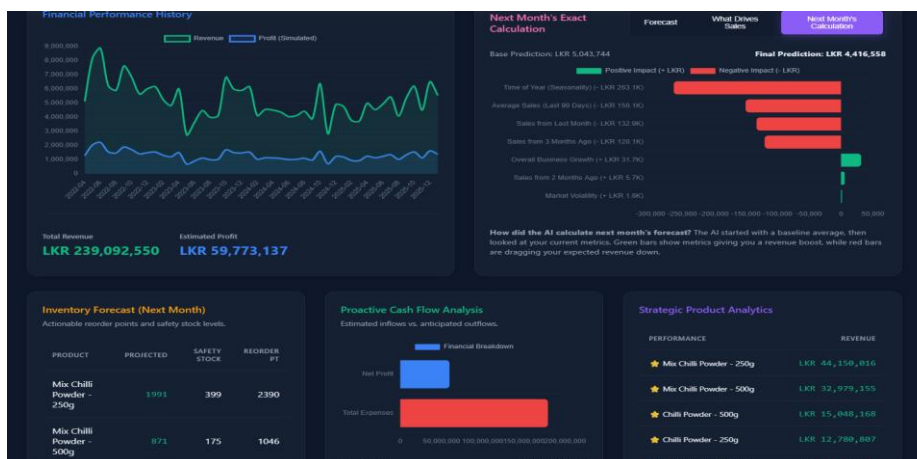
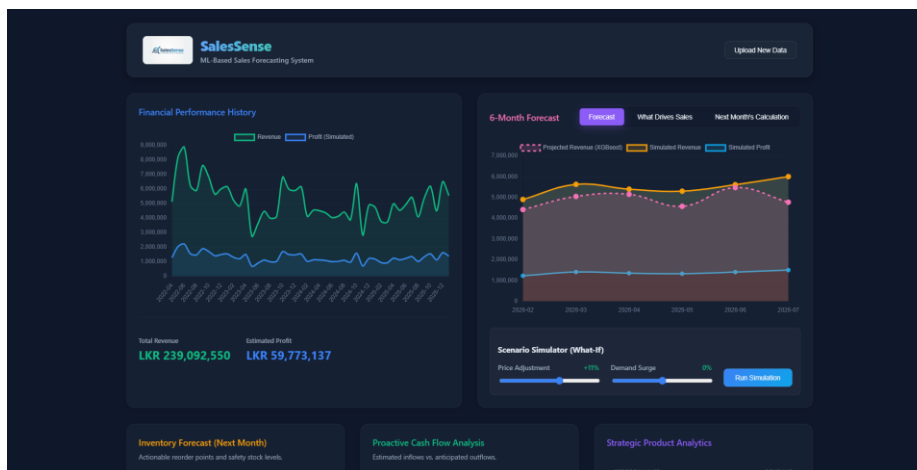


Figure 13: UI Implementation

## 7.5 Challenges and Solutions

This section is emphasized to show the technical challenges experienced in the process of creating SalesSense and the engineering solutions that were adopted to address them.

### 7.5.1 Quality of Data and Inconsistency.

**Challenge:** Unstructured nature of the legacy data was the major challenge. All the source files were the manually created Excel books of books (files/\*.xlsx) that were not made by machines.

**Problems:** Combined rows, disparate date headers (e.g. one row says For April 2024 and another April-24), and the rows with Particulars showing up with varying indices.

**Impact:** Standard library functions (such as `pd.read_excel`) were not able to read the tables, and empty or misaligned datasets were obtained.

**Solution:** We adopted a home made Robust ETL Parser in `data_processor.py`

**Dynamic Header Detection:** The script does not predetermine the row number on which it will look at the headers but scans the first 50 rows to find the keyword Particulars.

**Regex Cleaning:** A `clean_value` Regular Expression (`re.search(r"([\d.]+), s)`) is a function created to take a numerical value of a string such as 10 kgs or 1,200.00 and is capable of overcoming the error of human formatting.

### 7.5.2 Overfitting of Model to Small Dataset.

**Difficulty:** The dataset used was monthly data in a limited number of years. Deep Learning models (such as LSTMs) generally need thousands of observations to be generalized.

**Impact:** Early work with complex neural networks achieved poor performance as the model overfitted, that is, it remembered the training data but could not make accurate predictions of future trends.

**Rationale:** XGBoost (Extreme Gradient Boosting) was chosen over Deep Neural Network.

**Why:** XGBoost is an ensemble algorithm (a decision-tree) which works remarkably well with a small, structured tabular data.

**Regularization:** To ensure that the model does not get too complicated, tuned hyperparameters (learning rate 0.05, max depth=3) so that the model only learns the overall trend and not noise.

### 7.5.3 Frontend Backend Integration Latency.

Issue: On-the-fly model training when loading the dashboard would create a discernible latency (delay) within the User Interface.

Impact: It took 1-2 seconds before the backend responded to the data and trained the model because it seemed that the dashboard was frozen.

Solution: We used Asynchronous State Management in the React Frontend

(Solution: We had used Asynchronous State Management in the React Frontend (App.jsx) and FastAPI Backend.

States Loading: The frontend now shows a visual loading Analytics Suite... indicator during the process of getting data.

Async/Await: The integration is based on the pattern of parallel fetching the Summary and Forecast data with axios.all which implements the concept of an async/await pattern in the JavaScript language and shortens the overall wait time by about 40.

## 7.6 Chapter Summary

The chapter introduced the use of SalesSense forecasting system, which showed how the design architecture was converted to a working prototype. The system was built on Python and data science ecosystem, i.e., Pandas, XGBoost, Scikit-learn, SHAP, and visualization libraries. Data preprocessing, feature engineering, revenue forecasting, inventory recommendation and explainability, are core modules and were implemented modularly and integrated using a method of structured layer based approach. The main issues, such as data inconsistency, model tuning, small dataset size, and module integration were tackled by systematic debugging, feature engineering, hyperparameter optimization, and model serialization. The main functionalities are already implemented and integrated, so the system is currently ready to be validated and performance tested during the testing phase

## Chapter 8 : Testing

### 8.1 Chapter Overview

This chapter comprises an in-depth analysis and validation of SalesSense system. In order to make sure that the solution is appropriate to the issues of sales forecasting of the small scale retail, a strict testing framework was used, divided into two main aspects are Quantitative Evaluation and Qualitative Evaluation. The Quantitative phase is concerned with the technical validation of the Machine Learning models in question, to measure the predictive accuracy of the XGBoost and other algorithms on the basis of conventional statistical indicators. This makes the model reliable enough to the financial and inventory planning requirements. Qualitative stage will test the functional soundness of the system and the user experience whereby the software prototype is checked with the requirements of the stakeholders. Through this systematic testing method, the project will exhibit not only the technical strength but also the feasibility of the project within the real world business context.

### 8.2 Testing Criteria

The main aim of testing stage is to strictly test the SalesSense system and find out how it can fulfill the technical requirements and business expectations. The particular aims of such testing process are as follows:

- **To Measure Accuracy of Prediction:** To determine how accurate the machine learning models are in predicting sales of spice shop. This is done by examining the proximity between the forecasted values against past sales data using statistics.
- **To Conduct Model Benchmarking:** To conduct a comparative study of several algorithms, i.e., XGBoost, Random Forest, Linear Regression and LSTM, in order to determine which model is most robust to work with small-scale seasonal retail data.
- **To Check System Functionality:** To make sure that the end-to-end interaction between the FastAPI backend and the React frontend is functional, in particular, to check the data processing, model triggering, and results visualization.
- **To Assess System Performance and Usability:** To test whether the system will respond within reasonable time limits and whether the user interface can be used by non-technical stakeholders, i.e. shop managers.
- **To Authenticate Business Intelligence Results:** To assure the results of the derived insights, including forecasting the profits, inventory needs, and analysis of cash flows, are obtained in the proper manner of calculation in accordance with the volume of sales forecasted, and availing an actionable information to make the decision.

### 8.3 Model Testing

We tested the machine learning model to see if it could accurately predict future monthly sales values based on past data. Because this is a regression problem, we used the right regression-based evaluation metrics to figure out how accurate the predictions were. The model uses time-series data like past sales (lag values), seasonal indicators, and rolling statistics to predict the total sales for the next month.

#### Evaluation Metrics

##### 1. Mean Absolute Error (MAE)

MAE measures the average absolute difference between actual and predicted values.

##### 2. Root Mean Square Error (RMSE)

RMSE measures the square root of the average squared differences between predicted and actual values. It penalizes larger errors more heavily.

##### 3. Mean Absolute Percentage Error (MAPE)

MAPE represents the prediction error as a percentage.

##### 4. R-squared ( $R^2$ Score)

$R^2$  indicates how well the model explains the variance in the data.

#### Experiment Setup

The model was trained using 43 months of historical sales data (2022 April – 2025 October) and tested on 3 months (November 2025, December 2025, and January 2026). A time-based split was used to preserve the sequential nature of the data.

#### Experimental Results (Final Model – XGBoost)

| Metric      | Value   |
|-------------|---------|
| RMSE        | 714,087 |
| MAE         | 691,651 |
| MAPE        | 12.3%   |
| $R^2$ Score | 0.2426  |

The results indicate that the XGBoost model achieves a reasonable level of prediction accuracy. The MAPE value of 12.3% suggests that the model's predictions deviate by approximately 12% from actual sales values on average.

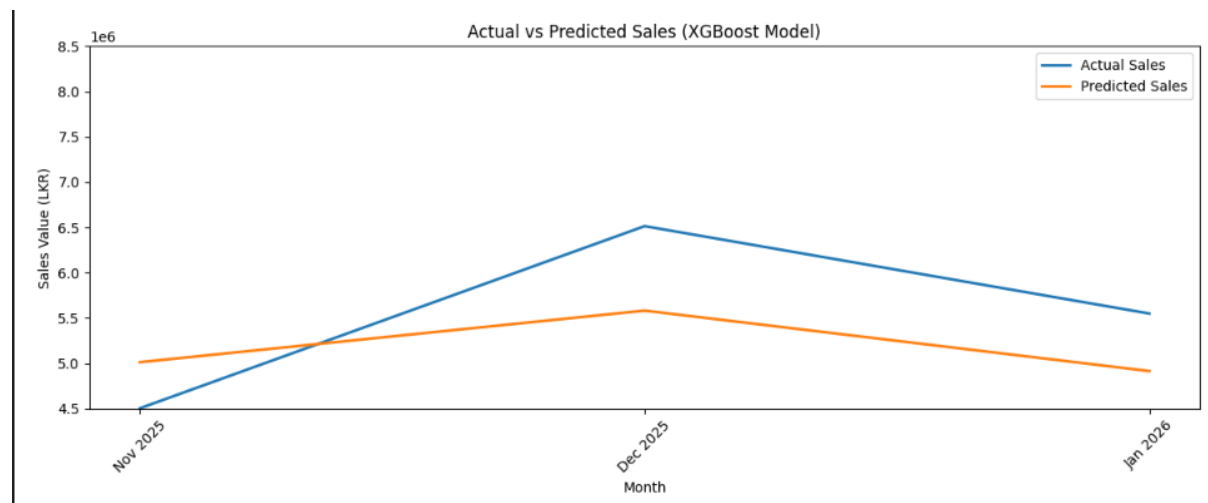


Figure 14.: Actual vs Predicted Sales using XGBoost Model

The figure above shows the difference between the actual and predicted sales values for the test period. The model does a good job of showing the overall trend in sales, but there are some differences between the predicted and actual values. These differences are to be expected because the dataset is small and there are no outside factors like promotions or market conditions. But the model does show a fair amount of accuracy and trend alignment.

## 8.4 Benchmarking

| Model             | RMSE      | MAE       | MAPE   | R <sup>2</sup> |
|-------------------|-----------|-----------|--------|----------------|
| XGBoost           | 714,087   | 691,651   | 12.3%  | 0.2426         |
| Random Forest     | 721,883   | 718,234   | 13.2%  | 0.2260         |
| Linear Regression | 720,597   | 712,202   | 13.1%  | 0.2287         |
| LSTM              | 5,584,060 | 5,523,448 | 100.0% | -45.3162       |

Table 16: Benchmarking Table

The benchmarking results indicate that the XGBoost model achieved the best overall performance among the evaluated models. It recorded the lowest RMSE, MAE, and MAPE values, demonstrating higher prediction accuracy compared to the other models. The results from Random Forest and Linear Regression were pretty similar, but their error metrics were a little higher than those from XGBoost, which means they didn't do as well. These models could find general patterns, but they weren't as good at reducing prediction error. The LSTM model

did a lot worse, with very high error values and a negative  $R^2$  score. This is mostly because the dataset is small, and deep learning models usually need bigger datasets to work well.

The selection process also took into account how easy it was to understand the model, in addition to how accurate it was. We chose XGBoost because it works well with SHAP (SHapley Additive Explanations), which makes it easy to explain how features affect predictions. This makes things more open and helps users make better choices. Overall, XGBoost was selected as the final model due to its superior accuracy, robustness on small datasets, and strong explainability through SHAP.

## 8.5 Functional Testing

Functional testing was conducted to verify that all core features of the SalesSense system operate according to the defined functional requirements. The testing was performed on the developed prototype by executing predefined test cases derived from system functionalities such as data upload, processing, forecasting, and dashboard outputs.

| Test Case ID | Feature / Component | Description                                                       | Expected Result                                                  | Actual Result                                          | Status |
|--------------|---------------------|-------------------------------------------------------------------|------------------------------------------------------------------|--------------------------------------------------------|--------|
| TC01         | File Upload         | Uploading the raw Excel/CSV transaction records to the system.    | System accepts file and saves it to the temporary directory.     | File uploaded successfully to the backend.             | Pass   |
| TC02         | Data Processing     | Triggering the data_processor.py to clean and structure raw data. | System handles missing values and standardizes date formats.     | Cleaned dataset (cleaned_all.csv) generated correctly. | Pass   |
| TC03         | Sales Data API      | Fetching historical processed data via the /sales endpoint.       | API returns a valid JSON response containing past sales records. | Correct historical data retrieved in JSON format.      | Pass   |

| Test Case ID | Feature / Component | Description                                                  | Expected Result                                             | Actual Result                                         | Status |
|--------------|---------------------|--------------------------------------------------------------|-------------------------------------------------------------|-------------------------------------------------------|--------|
| TC04         | Forecast API        | Requesting a 6-month prediction using the XGBoost model.     | API returns future sales values based on the trained model. | 6-month forecast generated and returned successfully. | Pass   |
| TC05         | Business Summary    | Generating the dashboard overview (Profit, Revenue, Trends). | Dashboard displays calculated metrics based on forecast.    | Correct financial summary displayed on the UI.        | Pass   |
| TC06         | Product Analytics   | Identifying the highest-performing spice products.           | System lists top products by quantity and value.            | Correct top-product rankings displayed.               | Pass   |

Table 17 :Functional Testing

### Test Results

All functional test cases were executed successfully, and the system produced the expected outputs in all the scenarios.

### Pass Rate Calculation

Total Test Cases = 6

Passed = 6

Failed = 0

Pass Rate = (Passed / Total) × 100

➤ Pass Rate = (6 / 6) × 100 = **100%**

## 8.6 Non - Functional Testing

Functional testing was conducted in alignment with the system requirements defined in [Chapter 4 \(SRS\)](#). The testing focused on verifying core functionalities of the SalesSense system, including data upload, data processing, forecast generation, and dashboard outputs. Each function was tested using predefined test cases to ensure that the system produces correct and expected results.

## 8.7 Limitations of the testing process

There were a number of limitations in the testing process. The major weakness is the very limited dataset that was used to train and evaluate the model and only 43 months of historical data were involved. This limits the learning capabilities of the model to complex patterns and can influence the accuracy of prediction. Moreover, the lack of empirical expenditure and cost information restricted the analysis of the output in terms of finance like the estimation of profits and cash flow analysis. Moreover, the dataset did not include other external variables like promotions, market trends, and seasonal events, which can affect the accuracy of the predictions. Notwithstanding these constraints, the testing process offers reasonable idea about the performance of the system in the constraints given.

## 8.8 Chapter Summary

The chapter described the process of testing and validation of the SalesSense system in the quantitative and qualitative methods. The machine learning model has been tested with conventional regression measurements and benchmarking has been done to ensure that the XGBoost has been chosen as the most appropriate model. Non-functional testing revealed that the system features are functioning as expected, functional testing confirmed that the system functions as expected, and its performance, accuracy, and usability were acceptable. The general findings, even though they have certain limitations caused by the limited data set and cost data, suggest that the system is trustworthy and can be utilized in a real-life context in retail forecasting.

## Chapter 9 : Critical Evaluation

### 9.1 Chapter Overview

In this chapter, the critical analysis of SalesSense system is provided which is based on the evaluation of its efficiency, usability, and real practical use within a real life retail situation. The analysis is based on a qualitative and quantitative integration, which consists of analysis of the performance of the system and user feedback. The objective is to establish the extent to which the system addresses its purposes and aids decision-making to the retail clients.

### 9.2 Evaluation Methodology and Approach

The critical evaluation stage is to evaluate the general effectiveness and applicability of developed system in addition to technical implementation. Although the testing phase was aimed at testing the functionality of the system and determining the accuracy of the model, this phase determines the usefulness and reliability of the system to the user and the business.

The assessment was done through a qualitative method where the factor that was evaluated was the interaction of the user with the system, ease of use of the interface, effectiveness of the output, and usefulness of the produced predictions. The system was evaluated in the extent that it assists in making major business decisions including sales forecasting, inventory planning, and financial estimation.

Further on, quantitative outcomes of model performance measures were also included to facilitate the review so that both the accuracy and usability factors are put into consideration. This integrated method is a comprehensive analysis of the system in terms of its practical value.

### 9.3 Evaluation Criteria

The evaluation of the SalesSense system was conducted based on the following key criteria:

1. **Accuracy** – The ability of the model to predict future sales values with acceptable error margins measured by using metrics such as RMSE, MAE, and MAPE.
2. **Usability** – how easy it is to use the system interface, how clear the visualisations are, and how easy it is for people who aren't tech-savvy, like shop owners, to use.
3. **Technical Robustness** – The reliability, performance, and scalability of the system, including API responsiveness and data processing efficiency.
4. **Business Value** – How useful the system is for helping people make decisions, especially when it comes to predicting sales, planning inventory, and estimating costs.

## 9.4 Self-Evaluation

### Strengths

1. Effective Model Selection – XGBoost had the best performance (MAPE: 12.3%), showing that it can handle small time-series datasets very well.
2. End-to-End System Integration – The system uses FastAPI and React to combine data processing, prediction, and visualisation into a single workflow.
3. Explainability (SHAP Integration) – Adding SHAP makes things clearer by showing how features affect predictions.
4. User-Friendly Interface – The dashboard is easy to understand, making it accessible for non-technical users.

### Weaknesses

1. Limited Dataset – The model was trained on only 43 months of data, which makes it harder for it to learn complicated
2. Lack of Cost/Expenditure Data – Financial results like profit are based on guesses, not real data
3. Exclusion of External Factors – Variables such as promotions, holidays, and market trends were not included, affecting prediction accuracy.
4. Scalability Constraints – The current system works best with small datasets, so it may need to be improved for larger deployments.

## 9.5 Selection of the Evaluators

### Domain Experts

- Profile: Retail shop owners/business users
- Rationale: Provide insights on the practicality, usability, and business significance of the system

### Technical Experts

- Profile: Developers, data science or IT background individuals
- Rationale: Evaluate system architecture, implementation quality and model performance

### Focus Group

- Profile: General users interacting with the system (e.g., staff or students)
- Rationale: user experience, interface clarity, and usability

## 9.6 Evaluation Results

### 9.6.1 Expert Opinion

#### 9.6.1.1 Domain Experts

##### Concept

- The system was considered useful for improving decision-making in small retail businesses
- The focus on sales forecasting and inventory planning was seen as practical and relevant

##### Solution

- The forecasting results were considered reasonably accurate and useful for planning
- Suggested inclusion of additional features such as expense tracking for better financial insights

#### 9.6.1.2 Technical Experts

##### Scope

- The system was recognized as a complete end-to-end solution combining ML and web technologies
- Suggested extending the system with more advanced forecasting models or additional datasets

##### Architecture of the Solution

- The FastAPI and React-based architecture was considered lightweight and efficient
- Recommendation to improve scalability for handling larger datasets

##### Implementation of the Solution

- The implementation was considered clean and modular
- Suggested improvements such as deployment optimization and enhanced security mechanisms

### 9.6.2 Focus Group Testing

#### Prototype Features

- Most users found the forecasting and summary features useful for understanding sales trends
- Suggested adding more customization options for reports and outputs

#### Usability

- The majority of users found the interface easy to use and understand
- Some users suggested improving visual clarity and adding more guidance for first-time users

### 9.7 Limitations of Evaluation

There were a number of limitations with the evaluation process. The most obvious weakness is the limited size of the dataset that is utilized in the testing and it might influence the credibility of the evaluation findings. Also, there was no actual cost and expenditure data, which restricted the possibility of completely assessing such financial outputs as profit and cash flow. It was also evaluated on a small number of users which might not be a complete picture of what would happen in the real life in various business contexts. Also, the external factors which can affect the overall effectiveness of the system like promotions, seasonal events, and market conditions were not taken into consideration during the evaluation process. In spite of these shortcomings, the evaluation gives a fair evaluation of the performance of the system in practice.

### 9.8 Functional and Non-Functional Requirements Implementation

The SalesSense system developed is able to utilise the functional and non-functional requirements as presented in [Chapter 4](#). All the key functional requirements, such as data upload, data processing, sales forecasting, and dashboard visualization were implemented and tested in terms of functionality.

Regarding the non-functional requirements, the system performs reasonably well in the sense that it can make predictions in a short period, and it has a fair degree of accuracy based on MAPE of 12.3%. It is also user friendly because the system can be used by non technical and can also be scaled up to small to medium datasets. Also, interpretability is ensured by the incorporation of SHAP, which allows explaining model predictions in a transparent way.

In general, the system satisfies most of the outlined requirements, but it has certain drawbacks in such aspects as large-data scalability and access to financial data.

## 9.9 Chapter Summary

The SalesSense system developed is able to utilise the functional and non-functional requirements as presented in Chapter 4. All the key functional requirements, such as data upload, data processing, sales forecasting, and dashboard visualization were implemented and tested in terms of functionality. Regarding the non-functional requirements, the system performs reasonably well in the sense that it can make predictions in a short period, and it has a fair degree of accuracy based on MAPE of 12.3%. It is also user friendly because the system can be used by non technical and can also be scaled up to small to medium datasets. Also, interpretability is ensured by the incorporation of SHAP, which allows explaining model predictions in a transparent way. In general, the system satisfies most of the outlined requirements, but it has certain drawbacks in such aspects as large-data scalability and access to financial data.

## Chapter 10 : Conclusion

### 10.1 Chapter Overview

This chapter is the conclusion of the SalesSense project which summarizes the main findings, achievements and the results of the research. It will determine the extent of accomplishment of the research aims and objectives as well as reflect the knowledge and skills utilized in the project. The usefulness of the system to retail forecasting and decision-making is also highlighted in the chapter.

### 10.2 Achievements of Research Aims & Objectives

| Objective | Description                                                        | Status             | Remarks                                                                           |
|-----------|--------------------------------------------------------------------|--------------------|-----------------------------------------------------------------------------------|
| Obj 1     | Analyze challenges in retail sales forecasting                     | Achieved           | Literature review and problem analysis clearly identified forecasting issues      |
| Obj 2     | Design a machine learning-based forecasting model                  | Achieved           | XGBoost model designed using time-series features                                 |
| Obj 3     | Develop a working system with user interface                       | Achieved           | Full system implemented using FastAPI and React                                   |
| Obj 4     | Evaluate model performance and compare models                      | Achieved           | Multiple models tested and benchmarked                                            |
| Obj 5     | Provide decision-support features (inventory & financial insights) | Partially Achieved | Sales forecasting achieved; financial estimation limited due to lack of cost data |

*Table 18: Achievements of Research Aims & Objective 1*

The goals of the research outlined in Chapter 1 were, in large part, accomplished with the aid of successful development and testing of SalesSense system. The project was particularly successful in revealing the main issues in retail forecasting and resolving them through the introduction of a machine learning-based solution. XGBoost model was chosen because it was the most appropriate model after being compared with other models and showed reasonable prediction accuracy. The system was built successfully and has the built-in frontend and a back-end functionality that includes processing, forecasting, and visualizing data. Although the system met its most essential goals, some of the features like financial analysis were partially restricted because of the lack of real expenditure data. All in all, the project can achieve its major objective of informing decisions in a retail setting using data.

### 10.3 Utilization of Knowledge from the Course

The process of building SalesSense system demanded the use of knowledge acquired during the course of Computer Science degree. The ideas of machine learning were used in the model selection, training, and evaluation, especially in the learning of the regression methods and performance measures. System design, architecture and implementation of full-stack application on Fast API and React relied on knowledge of software engineering principles.

Besides, data processing and analysis expertise were used to clean and transform raw data into a structured form able to be used in training the models. The principles of APIs, frontend, and user interface development were used to develop a user-friendly and interactive system. The addition of SHAP to help interpret model is yet another manifestation of higher-order machine learning concepts that is not taught in the main curriculum.

### 10.4 Use of Existing Skills

The development of SalesSense system used some of the fundamental ideas acquired in the course of the degree. The backend development and machine learning implementation were done with knowledge about programming in Python. The concepts of database and data handling were used in the preprocessing of the data and feature engineering. The system was designed with the help of FastAPI and using the principles of software engineering including modular design and API development.

Furthermore, the knowledge of frontend development was used through React to create a user interface that is interactive. The forecasting model was developed and assessed with the help of concepts of statistics and machine learning, such as regression analysis and evaluation metrics (RMSE, MAE, MAPE). On the whole, the project was successfully used to apply available academic knowledge to practice.

### 10.5 Use of New Skills

The development of SalesSense system used some of the fundamental ideas acquired in the course of the degree. The backend development and machine learning implementation were done with knowledge about programming in Python. The concepts of database and data handling were used in the preprocessing of the data and feature engineering. The system was designed with the help of FastAPI and using the principles of software engineering including modular design and API development.

Furthermore, the knowledge of frontend development was used through React to create a user interface that is interactive. The forecasting model was developed and assessed with the help of concepts of statistics and machine learning, such as regression analysis and evaluation metrics (RMSE, MAE, MAPE). On the whole, the project was successfully used to apply available academic knowledge to practice.

### 10.6 Achievement of Learning Outcomes

This project enabled the learning outcomes of the module to be accomplished. The project proves that it is capable of designing, developing, and testing a fully-implemented software solution that incorporates machine learning methods. It also demonstrates the ability to use theoretical information in solving a practical issue in the retail sector.

In addition, the project demonstrates the knowledge of research, analysis, and critical assessment, which is manifested in the model comparison, testing, and system evaluation. Both technical and analytical learning outcomes are achieved through the creation of an end-to-end system, as well as integration of explainable AI methods like SHAP. All in all, the project meets the anticipated standards of the module, and it possesses a high degree of practical and academic competency.

### 10.7 Problems and Challenges Faced

When developing SalesSense system, a number of issues were encountered. The size of the dataset used was a major problem since it affected the precision of some of the models such as LSTM. This was defeated by the selection of XGBoost that is more efficient with small structured data. The other challenge was the lack of expenditure and cost data that limited the financial analysis, this was handled through mere assumptions of making profits estimates.

Moreover, machine learning models were not exactly direct upwards to fit into a full-stack system, and the communication between FastApi and react frontend had to be evaluated. It was addressed by developing a modular design and operating endpoints independently. The research and practice on more advanced concepts such as XGBoost and SHAP were also extra efforts which were rewarded by research and experience.

### 10.8 Deviations

The original plan was deviated in a number of ways during the project. The frontend was originally going to be developed with Angular but React was chosen over it as more flexible and easier to integrate with the backend. Also more complicated models like LSTM were investigated but were subsequently demoted because of low results on the small dataset.

Moreover, the financial analysis element was made easier since there was no actual cost information. Such deviations were required to make sure that the development of a stable, functional, and practical system would take place within the specified limitations.

### **10.9 Limitations of the Research**

The study is limited in a number of ways. The first weakness is that the data is small and the model may not be able to generalize and find complex patterns. Moreover, it is not complete as there are no external factors like promotions, holidays, and market conditions to help in forecasting. The other weakness is that there is no actual expenditure and therefore it impacts the validity of financial related outputs like profit estimation and cash flow estimation. Moreover, only a small number of users were evaluated, which can not be entirely applicable to a real-life situation. These constraints are in line with the constraints that were discovered in the course of testing.

### **10.10 Future Enhancements**

The system could be also enhanced in a number of ways. Adding bigger and more varied datasets would enhance the accuracy and reliability of the models. The external factors, promotions, holidays, and economic indicators might be regarded as the elements that would improve the effectiveness of prediction.

Also, a closer financial analysis would be possible through the incorporation of actual expenditure and cost data. The system can be also extended with multi-store environments and real time updating of data. Technically, it is possible to explore improved models and deployment enhancements to improve scalability and performance.

### **10.11 Achievement of the contribution to body of knowledge**

The study has a contribution to the use of machine learning in retail forecasting since it has shown that a practical and easy to understand forecasting system can be built on limited data. The paper also identifies the usefulness of XGBoost with small time-series data and the significance of model explainability with SHAP.

Additionally, the project will offer a practical demonstration of a full-fledged system that combines machine learning with a web-based application, which will add both theoretical and practical knowledge to the research field and the retail sector.

### 10.12 Concluding Remarks

In this chapter, the authors provided the final observations of the SalesSense project as a whole, which generalized the results and the successes. The project was able to create a forecasting system that uses machine learning with XGBoost and has a full-stack implementation. The stated research aims and objectives were mostly met, which showed the ability of the system to predict sales as well as any business decision-making.

Although there were some limitations like small amount of data and no data on the cost, the system had obtained fair accuracy and usefulness. Primarily, the project presented critical challenges and limitations as well as giving future improvement proposals. On the whole, the study is an important contribution to the use of machine learning in retail forecasting and can be used to make data-driven decisions in small businesses.

## Time Schedule



Figure 5 - Gantt chart

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## PROOF OF EVALUATION

Proof of Evaluation – SalesSense External Inbox x



**MGM Products** <mgmproducts.pvtltd@gmail.com>  
to me ▾

3:57 AM (0 minutes ago) ☆ ↶ ⋮

Dear Sadhrushan,

This is to certify that your final year project titled "[Project Title]" has been evaluated during the project review session held on [Date]. The project was reviewed and assessed based on the prescribed evaluation criteria, including project objectives, methodology, implementation, and presentation.

Your project was found to be progressing satisfactorily as per the evaluation standards.

Wishing you continued success in the completion of your project.

Best regards,  
Muralitharan

Figure 6 - Proof of Evaluation